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CHAPTER 4

MONEY AND BANKING(CONTINUE)

SUGGESTED TIME: 7 PERIODS

IN THIS CHAPTER

This chapter completes the study of money and banking by examining how commercial banks operate and create credit, how the central bank (BEAC) regulates the monetary system, and how microfinance institutions extend financial services to low-income Cameroonians. It combines theoretical lessons with practical case studies rooted in the Cameroonian and CEMAC context, preparing students for data-response and essay questions at the GCE Advanced Level.

- By the end of this chapter I can draw and interpret the balance sheet of a commercial bank, distinguishing assets from liabilities.
- By the end of this chapter I can explain the mechanism of credit creation by commercial banks and compute the credit multiplier.
- By the end of this chapter I can identify and discuss the problems faced by commercial banks in Cameroon.
- By the end of this chapter I can define a central bank and state its functions and instruments of monetary policy.
- By the end of this chapter I can analyse the role of BEAC as the central bank of the CEMAC zone, including Cameroon.
- By the end of this chapter I can describe microfinance activities and their importance for financial inclusion in Cameroon.

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2. Section 2: Credit Creation by Commercial Banks and Their Problems in Cameroon (TU 30 — Practical Work 20; TU 31 — Further Studies 5)
3. Section 3: The Central Bank and the Case of BEAC (TU 32 — Lesson 92; TU 33 — Guided Work 5)
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CHAPTER OBJECTIVES

- By the end of this chapter I can draw and interpret the balance sheet of a commercial bank, distinguishing assets from liabilities.
- By the end of this chapter I can explain the mechanism of credit creation by commercial banks and compute the credit multiplier.
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- By the end of this chapter I can analyse the role of BEAC as the central bank of the CEMAC zone, including Cameroon.
- By the end of this chapter I can describe microfinance activities and their importance for financial inclusion in Cameroon.
- By the end of this chapter I can outline the elements of a good appraisal of a loan file by a credit union.
- By the end of this chapter I can solve GCE-style numerical and essay questions on banking and monetary policy.

PREREQUISITES

- Definition, functions and qualities of money (Lower Sixth / ECO03)
- Types of money: fiduciary and scriptural (bank) money
- The demand for and supply of money; quantity theory basics
- Inflation: meaning, causes and measurement in Cameroon
- Basic arithmetic with percentages and ratios

1 SECTION 1: THE BALANCE SHEET OF A COMMERCIAL BANK (TU 29 — LESSON 91)

Aminatou, a trader at Mokolo Market in Yaoundé, deposits her daily sales of 150 000 FCFA at her commercial bank every evening before going home. Two weeks later, her neighbour obtains a loan of 1 000 000 FCFA from the very same bank to buy a taxi. Where did the bank find that money? Meanwhile, the BEAC in Yaoundé announces a rise in its key interest rate, and suddenly loans become more expensive across the whole CEMAC zone, while the credit union in her quarter refuses a loan file that seemed convincing. Behind these everyday scenes lie the **balance sheet** of commercial banks, the mysterious power of credit creation, the supervisory role of the central bank and the growing world of microfinance. This chapter opens the doors of the banking system so that you can explain, calculate and evaluate what really happens with money in Cameroon.

Our first problem is simple to state but fundamental: how can we describe, at a glance, everything a bank owns and everything it owes? The answer is the balance sheet.

1.1 Recall: What Is a Commercial Bank?

Before opening the accounts, let us recall what a commercial bank actually does. When Aminatou hands her 150 000 FCFA to the cashier, the bank is performing its first function: **deposit-taking**. When her neighbour drives away in his new taxi, the bank has performed its second function: **lending**. And when Aminatou pays her wholesaler by cheque or by mobile transfer instead of carrying cash across town, the bank is performing its third function: providing **payment services**.

DEFINITION **Commercial bank**

A **commercial bank** is a financial institution whose main activities are (i) collecting deposits from the public, (ii) granting loans and overdrafts to households and firms, and (iii) providing means of payment (cheques, transfers, bank cards, mobile banking). Its objective is to make a profit from the difference between the interest it charges on loans and the interest it pays on deposits — this difference is called the **interest margin** or spread.

In Cameroon, well-known commercial banks include **Afriland First Bank**, **Société Générale Cameroun (SGBC)**, **BICEC** (Banque Internationale du Cameroun pour l'Épargne et le Crédit) and **Ecobank Cameroun**. All of them are licensed and supervised by the regional central bank, the **BEAC** (Bank of Central African States), which we shall study in Section 3.

1.2 The Meaning and Structure of a Balance Sheet

Imagine you freeze time at midnight on 31 December and list, on one sheet of paper, everything the bank owns and everything the bank owes. That sheet is the balance sheet.

DEFINITION **Balance sheet**

A **balance sheet** is a financial statement that shows, at a given date, the **assets** of a bank (what it owns or is owed), its **liabilities** (what it owes to others) and its **net worth** (the residual belonging to shareholders).

DEFINITION **Assets and liabilities**

- An **asset** is anything the bank owns or anything owed to the bank (cash in the vault, loans granted to customers, buildings, government securities).
- A **liability** is anything the bank owes to others (customers' deposits, money borrowed from the central bank or from other banks).
- The **net worth** (or equity, or own funds) is what would remain for the shareholders if all liabilities were repaid: share capital plus accumulated reserves and retained profits.

PROPERTY **The accounting identity**

In every balance sheet, by construction:

$$\text{Total Assets} = \text{Total Liabilities} + \text{Net Worth}.$$

This identity always holds because net worth is defined as the residual: Net Worth = Assets − Liabilities. A balance sheet that does not balance contains an error. (The identity itself is definitional and does not require a proof in the examination, but you must be able to use it, for example to find a missing net worth figure.)

METHOD **Classifying any balance sheet item**

To decide where an item belongs, ask yourself one question:

1. Does the bank own it, or is it owed to the bank? → it is an asset (e.g. a loan granted to a customer: the customer owes the bank).
2. Does the bank owe it to someone else? → it is a liability (e.g. a customer's deposit: the bank owes the customer).
3. Is it the shareholders' own contribution or accumulated profit? → it is part of net worth.

WARNING **Deposits are NOT the bank's property**

The most common mistake at the GCE is to treat customers' deposits as something the bank owns. Wrong! When Aminatou deposits 150 000 FCFA, the bank must repay her on demand. The deposit is therefore a liability of the bank. Symmetrically, a loan granted to her neighbour is an asset of the bank, because the neighbour owes the money to the bank. Always reason from the bank's point of view.

1.3 The Liabilities Side: Where Does the Bank's Money Come From?

The liabilities side answers the question: who supplied the funds the bank is using? We distinguish four main sources.

- **Sight deposits (demand deposits).** Money placed on current accounts, withdrawable at any time by cheque, card or transfer. This is typically the largest liability of a commercial bank. Aminatou's deposit belongs here.
- **Time and savings deposits.** Money deposited for a fixed period or on savings accounts (e.g. a compte à terme at SGBC). The bank pays interest on them but can count on them more durably.
- **Borrowed funds.** Banks themselves borrow: either from the **central bank** (refinancing or rediscounting) or from other banks on the **interbank market**.
- **Shareholders' equity and reserves (net worth).** The capital contributed by the owners plus retained profits. This is the bank's safety cushion: it absorbs losses before depositors are affected.

1.4 The Assets Side: How Does the Bank Use the Funds?

The assets side answers the question: what has the bank done with the funds? We move from the most liquid to the least liquid (and generally most profitable) uses.

- **Cash in vault and reserves at the central bank.** Notes and coins in the tills, plus the account the bank holds at the BEAC. These earn little or nothing but allow the bank to meet withdrawals immediately. Part of these reserves is compulsory (required reserves imposed by the central bank).
- **Loans and overdrafts to customers.** Credits granted to households (the neighbour's taxi loan) and to firms (a loan to a cocoa exporter in Douala). This is usually the most profitable asset, but also the riskiest and least liquid: a loan cannot be recalled instantly.
- **Investment securities.** Treasury bills and government bonds bought by the bank. They earn interest and can be sold if the bank needs cash — a compromise between liquidity and profitability.
- **Fixed assets.** Buildings, branches, computers, vehicles. Necessary for operations but totally illiquid and not directly profitable.

ASSETS (uses of funds)	LIABILITIES & NET WORTH (sources)
Cash in vault and reserves at BEAC 800	Sight deposits 5 000
Loans and overdrafts to customers 6 200	Time and savings deposits .. 2 500
Investment securities (T-bills, bonds) 1500	Borrowed funds (BEAC, interbank) 500
Fixed assets (buildings, equipment) 500	Share capital and reserves .. 1000
Total assets 9 000	**Total liabilities + net worth 9 000**
Figures in millions of FCFA, simplified example	

Simplified balance sheet (T-account) of a Cameroonian commercial bank. Assets on the left equal liabilities plus net worth on the right: 9 000 = 9 000 million FCFA.

1.5 The Fundamental Trade-off: Liquidity, Profitability, Security

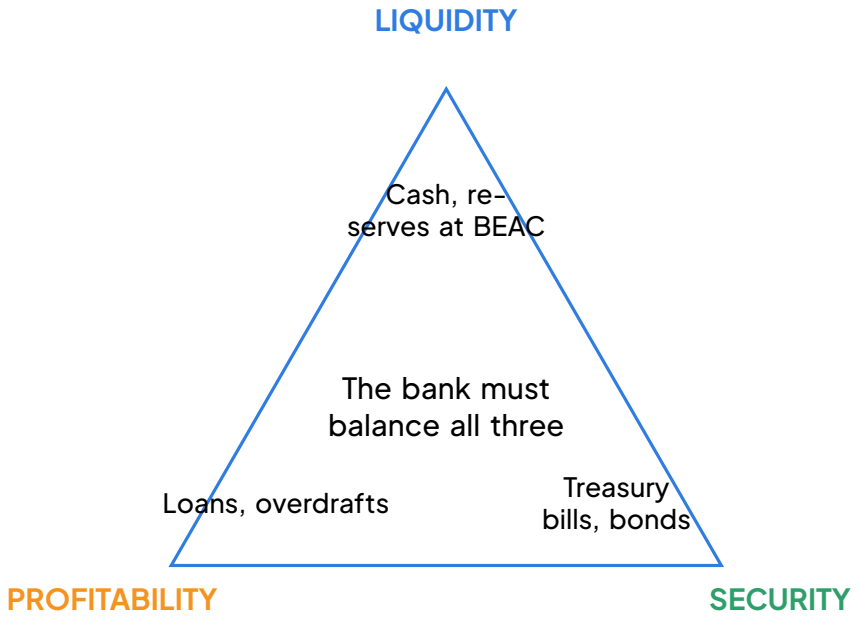
Why doesn't the bank lend out all of Aminatou's deposit and every other deposit, since loans earn the highest interest? Because depositors can come and withdraw their money at any time. If the bank has no cash left, it fails. Bank asset management is therefore a permanent balancing act between three objectives that pull in opposite directions.

DEFINITION **The three objectives of bank asset management**

- **Liquidity**: the ability to meet withdrawal demands immediately. Cash is perfectly liquid but earns nothing.
- **Profitability**: the search for the highest return. Long-term loans are the most profitable but the least liquid.
- **Security** (solvency): the avoidance of losses. Risky loans pay more but may never be repaid; safe assets (treasury bills) pay less.

PROPERTY **The liquidity–profitability–security trade-off**

For a bank, the more liquid and safe an asset is, the less profitable it tends to be; the more profitable an asset is, the less liquid and the riskier it tends to be. The bank must therefore diversify its assets: keep enough cash and reserves for daily withdrawals, hold some easily sellable securities, and lend the rest at profitable rates to reliable borrowers.



The liquidity–profitability–security triangle: moving towards one vertex means moving away from the other two.

1.6 Key Ratios Derived from the Balance Sheet

The balance sheet is not just a photograph; it is a diagnostic tool. Three ratios are essential at the GCE level.

DEFINITION **Key balance sheet ratios**

- Reserve ratio = $\frac{\text{Reserves (cash + deposits at central bank)}}{\text{Total deposits}}$. It measures the fraction of deposits kept liquid.
- Liquidity ratio = $\frac{\text{Liquid assets (cash, reserves, short-term securities)}}{\text{Short-term liabilities (sight deposits)}}$. It measures the ability to face withdrawals.
- Loan-to-deposit ratio = $\frac{\text{Total loans}}{\text{Total deposits}}$. A high value means aggressive lending; a value above 100% means the bank lends more than its deposits and must borrow elsewhere.

WORKED EXAMPLE 1 **Reading the balance sheet of "Mokolo Bank Ltd"**

Statement. Using the simplified balance sheet of Mokolo Bank Ltd (figures in millions of

FCFA): cash and reserves at BEAC = 800; loans = 6 200; securities = 1 500; fixed assets = 500; sight deposits = 5 000; time deposits = 2 500; borrowed funds = 500; capital and reserves = 1 000.

1. Verify the accounting identity.
2. Compute the reserve ratio, the liquidity ratio and the loan-to-deposit ratio.
3. Comment on the bank's policy.

SOLUTION

Solution.

1. Total assets = 800 + 6 200 + 1 500 + 500 = 9 000. Total liabilities + net worth = 5 000 + 2 500 + 500 + 1 000 = 9 000. The identity holds: 9 000 = 9 000.
2. Total deposits = 5 000 + 2 500 = 7 500.

$$\text{Reserve ratio} = \frac{800}{7\,500} \approx 10.7\%$$
$$\text{Liquidity ratio} = \frac{800 + 1\,500}{5\,000} = \frac{2\,300}{5\,000} = 46\%$$
$$\text{Loan-to-deposit ratio} = \frac{6\,200}{7\,500} \approx 82.7\%$$

3. The bank keeps about 10.7% of deposits as reserves and holds comfortable liquid assets against sight deposits (46%). Its loan-to-deposit ratio of 82.7% shows an active but not reckless lending policy: most deposits are transformed into profitable loans, while liquidity remains adequate. The bank respects the liquidity–profitability–security balance.

WARNING **Exam traps on ratios**

- Do not confuse the reserve ratio (reserves over deposits) with the capital ratio (net worth over assets).
- In the liquidity ratio, only short-term liabilities (sight deposits) appear in the denominator, not time deposits.
- Always express ratios as percentages and interpret them: a bare number without commentary earns only part of the marks.

EXERCISE 1 **Constructing a balance sheet**

Douala Star Bank presents the following items at 31 December (millions of FCFA): loans to customers 4 000; sight deposits 3 500; cash and reserves at BEAC 600; treasury bills 900; time deposits 1 800; buildings 300; borrowing from BEAC 200; share capital and reserves: to be determined.

1. Classify each item as asset, liability or net worth, and draw up the balance sheet.
2. Find the value of share capital and reserves using the accounting identity.
3. Compute the reserve ratio and the loan-to-deposit ratio, and comment briefly.

SOLUTION **Exercise 1**

1. Assets: loans 4 000; cash and reserves 600; treasury bills 900; buildings 300; total = 5 800. Liabilities: sight deposits 3 500; time deposits 1 800; borrowing from BEAC 200; total = 5 500. Net worth: unknown E.
2. E = 5 800 − 5 500 = 300 million FCFA.
3. Reserve ratio = 600/(3 500 + 1 800) = 600/5 300 ≈ 11.3%. Loan-to-deposit ratio = 4 000/5 300 ≈ 75.5%. The bank keeps a prudent liquidity cushion and lends about three quarters of its deposits: a balanced, moderately profitable policy.

KEY POINTS **Key points of Section 1**

- A commercial bank takes deposits, grants loans and provides payment services (Afriland, SGBC, BICEC, Ecobank in Cameroon).
- The **balance sheet** records assets (what the bank owns or is owed) and liabilities (what it owes) at a given date, with Assets = Liabilities + Net Worth.
- Deposits are liabilities; loans are assets — always reason from the bank's point of view.
- Bank management is a trade-off between **liquidity**, **profitability** and **security**.
- Master the reserve ratio, liquidity ratio and loan-to-deposit ratio, and always interpret your results.

DID YOU KNOW? **GCE exam tip**

At the GCE Advanced Level, examiners frequently give you a jumbled list of balance sheet

items and ask you to (i) construct the balance sheet, (ii) compute ratios and (iii) comment. Train yourself to classify each item in seconds with the question "does the bank own it or owe it?", and never forget that the two totals must be equal — if they are not, the missing figure is the net worth, found by subtraction.

Now that we can read a bank's accounts, a deeper mystery appears: if Mokolo Bank lends out deposits, and the taxi driver pays a car dealer who deposits the money again… can banks actually create money? That is the subject of Section 2.

2 SECTION 2: CREDIT CREATION BY COMMERCIAL BANKS AND THEIR PROBLEMS IN CAMEROON (TU 30 — PRACTICAL WORK 20; TU 31 — FURTHER STUDIES 5)

In Section 1 we studied the balance sheet of a commercial bank and saw that its main assets are reserves and loans, while its main liabilities are deposits. We also noticed something remarkable: a bank keeps only a fraction of its deposits as reserves and lends out the rest. This simple fact has an extraordinary consequence. When a bank grants a loan, the money does not disappear from the economy — the borrower spends it, the seller deposits it in another bank, that bank lends most of it again, and so on. One initial deposit can therefore give birth to a whole chain of new deposits. This process, called **credit creation**, is the subject of this section. We shall first explain how a single bank creates credit, then show how the whole banking system multiplies deposits, derive the **credit multiplier**, discuss the limits of the process, and finish with a practical numerical exercise and a study of the problems faced by commercial banks in Cameroon.

2.1 How a Single Bank Creates Credit

Imagine a trader in Douala who deposits 1 000 000 FCFA in cash at Bank A. If the central bank imposes a **required reserve ratio** of 10%, Bank A must keep 100 000 FCFA as reserves but is free to lend the remaining 900 000 FCFA, called its **excess reserves**. Suppose Bank A lends this 900 000 FCFA to a carpenter who uses it to buy timber. The timber seller receives the 900 000 FCFA and deposits it in her own bank, Bank B. Notice what has happened: the original 1 000 000 FCFA deposit still exists (the trader can withdraw it), and a new deposit of 900 000 FCFA has appeared at Bank B. Money has been created — not printed, but created through lending.

DEFINITION **Credit creation**

Credit creation is the process by which commercial banks generate new deposits (and therefore new money) by granting loans out of their **excess reserves**, that is, the part of deposits which exceeds the reserves they are legally or prudentially required to hold.

The key intuition is that a single bank can only lend its excess reserves — it cannot create money on its own beyond that, because the borrower will spend the loan and the bank will lose re-

serves to other banks. But the system as a whole can create a multiple of the initial deposit, because one bank's loan becomes another bank's deposit.

WARNING **A single bank versus the banking system**

A single bank in isolation can lend at most its excess reserves. It is the banking system as a whole — Bank A, Bank B, Bank C, … — that multiplies deposits. Never write that "one bank creates $1/r$ times the deposit"; that is the property of the system.

2.2 The Multiple Expansion of Deposits: A Step-by-Step Illustration

Let us follow the chain with a required reserve ratio r = 10% and an initial new deposit of 1 000 000 FCFA at Bank A. Each bank keeps 10% of the deposit it receives and lends out 90%.

Bank A. Receives a deposit of 1 000 000; keeps 100 000 as required reserves; lends 900 000.

Assets (Bank A)		Liabilities (Bank A)	
Reserves	100 000	Deposits	1 000 000
Loans	900 000		

Bank B. The 900 000 loan is spent and redeposited at Bank B. Bank B keeps 90 000 (10%) and lends 810 000.

Assets (Bank B)		Liabilities (Bank B)	
Reserves	90 000	Deposits	900 000
Loans	810 000		

Bank C. Receives 810 000; keeps 81 000; lends 729 000. The process continues through Bank D, Bank E, and so on, each deposit being 90% of the previous one.



The credit creation chain with a 10% reserve ratio: each bank keeps 10% and lends 90%, so deposits shrink at each round while their sum grows.

The deposits form a geometric progression:

$$1\,000\,000 + 900\,000 + 810\,000 + 729\,000 + \cdots = 1\,000\,000 \times \left(1 + 0.9 + 0.9^2 + 0.9^3 + \ldots\right).$$

The sum of an infinite geometric series with first term 1 and common ratio 0.9 is $\frac{1}{1-0.9} = \frac{1}{0.1} = 10$. Hence total deposits = 1 000 000 × 10 = 10 000 000 FCFA.

2.3 The Credit (Deposit) Multiplier

PROPERTY **The credit multiplier — formula**

If the required reserve ratio is r (expressed as a decimal) and banks hold no excess reserves while the public holds no extra cash, then the **credit multiplier** is

$M = \frac{1}{r}$, Total deposits = initial new deposit $\times \frac{1}{r}$, New credit created = Total deposits − initial deposit.

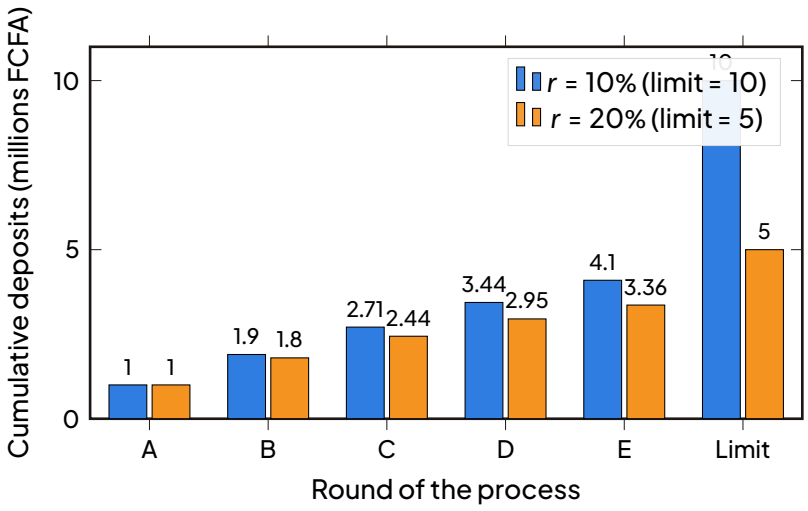
Proof sketch (instructive, not required at GCE): total deposits $= D_0\left(1 + (1-r) + (1-r)^2 + \dots\right) = D_0 \times \frac{1}{1-(1-r)} = \frac{D_0}{r}$.

EXEMPLE **Numerical applications in FCFA**

With an initial new deposit of 1 000 000 FCFA:

- If $r = 10\%$: $M = \frac{1}{0.1} = 10$; total deposits = 10 000 000 FCFA; new credit created = 9 000 000 FCFA.
- If $r = 20\%$: $M = \frac{1}{0.2} = 5$; total deposits = 5 000 000 FCFA; new credit created = 4 000 000 FCFA.
- If $r = 25\%$: $M = 4$; total deposits = 4 000 000 FCFA; new credit created = 3 000 000 FCFA.

The higher the reserve ratio, the smaller the multiplier: the central bank can therefore use r as an instrument of monetary policy.



Cumulative deposits created from an initial deposit of 1 000 000 FCFA. With r = 10% the total converges to 10 million FCFA; with r = 20% it converges to only 5 million FCFA.

METHOD **Computing total credit created**

1. Find the multiplier: $M = \frac{1}{r}$, where r is the reserve ratio as a decimal.
2. Multiply the initial new deposit by the multiplier to obtain total deposits.
3. Subtract the initial deposit from total deposits to obtain the newly created credit.

WARNING **Total deposits versus newly created credit**

A very common GCE mistake is to give total deposits as the answer when the question asks for credit created. The initial deposit is not created money — it existed before as cash. Newly created credit = total deposits − initial deposit. In our example, credit created is 9 000 000 FCFA, not 10 000 000 FCFA.

WARNING **The multiplier is a maximum**

The formula $M = 1/r$ assumes no cash leakages and no excess reserves. In Cameroon these leakages are large (many people hold cash outside banks, and banks often keep extra reserves), so the real credit creation is much smaller than the theoretical maximum.

2.4 Limits to Credit Creation

In reality, the expansion of deposits is limited by several factors:

- **Cash leakages (withdrawals).** If part of every loan is held by the public as cash instead of being redeposited — very common in Cameroon where many transactions are in cash — the amount returning to the banking system shrinks, and the multiplier falls.
- **Excess reserves.** Banks may prudently keep reserves above the legal minimum, especially when borrowers seem risky; idle excess reserves cannot create credit.
- **Lack of creditworthy borrowers.** Credit creation requires demand for loans from borrowers with viable projects and acceptable guarantees. If few firms or households qualify, banks cannot lend even when they have excess liquidity.
- **Central bank regulations.** The central bank can raise the reserve ratio, impose credit ceilings, or tighten refinancing conditions, directly limiting banks' ability to lend.

2.5 Practical Work: Credit Creation in the Cameroonian Banking System

WORKED EXAMPLE 2 **Computing total credit created**

Statement. A new deposit of 5 000 000 FCFA is made in a commercial bank in Yaoundé. The reserve requirement fixed by the central bank is 10%. Assuming no cash leakage and no excess reserves: (a) calculate the credit multiplier; (b) calculate the total deposits the banking system can support; (c) calculate the total credit newly created; (d) show the first three rounds of the process.

SOLUTION

Solution. (a) Multiplier: $M = \frac{1}{r} = \frac{1}{0{,}1} = 10$.
(b) Total deposits = 5 000 000 × 10 = 50 000 000 FCFA.
(c) Newly created credit = 50 000 000 − 5 000 000 = 45 000 000 FCFA.
(d) First three rounds:

Bank	Deposit (FCFA)	Reserves 10% (FCFA)	Loan (FCFA)
Bank A	5 000 000	500 000	4 500 000
Bank B	4 500 000	450 000	4 050 000
Bank C	4 050 000	405 000	3 645 000

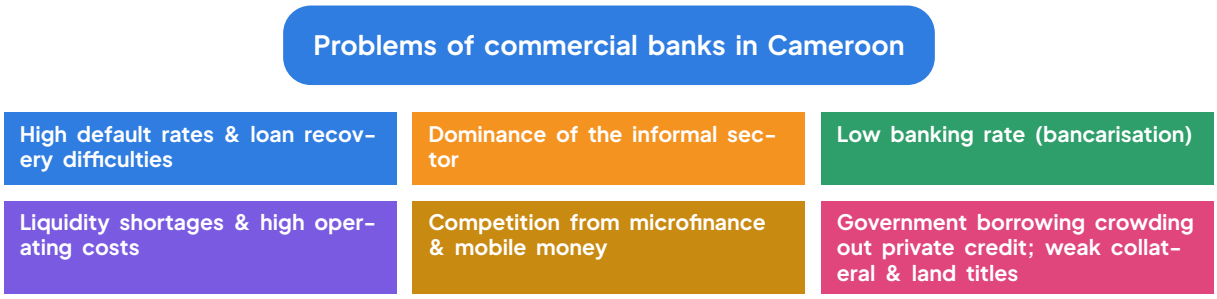
Each bank lends 90% of what it receives; the sum of all rounds converges to 50 000 000 FCFA of deposits, of which 45 000 000 FCFA is newly created credit.

METHOD **Exam tip (GCE)**

In the examination, show each step of the T-account chain (at least the first three banks), state the multiplier formula, and distinguish total deposits from credit created. Marks are awarded for the process, not only for the final figure.

2.6 Problems Faced by Commercial Banks in Cameroon (Further Studies)

Despite their theoretical power to create credit, commercial banks in Cameroon face serious practical difficulties that limit their role in financing the economy.



Mind map of the main problems faced by commercial banks in Cameroon.

High default rates and loan recovery difficulties. Many borrowers fail to repay their loans, and recovering debts through the courts is slow and costly. Non-performing loans tie up banks' funds and make them reluctant to lend.

Dominance of the informal sector. A large share of economic activity — small traders, artisans, transporters — operates informally, without proper accounts, tax identification or registered businesses. Banks cannot assess the risk of such borrowers and therefore refuse them credit.

Low banking rate (bancarisation). A large proportion of the population has no bank account, keeping savings in cash, in tontines or in microfinance institutions. This reduces the deposit base from which banks create credit.

Liquidity shortages and high operating costs. Banks sometimes face liquidity pressures, while the costs of branches, staff, security and energy are high relative to their business volume, pushing up interest rates on loans.

Competition from microfinance and mobile money. Credit unions, savings cooperatives and mobile money services (which allow transfers and payments by telephone) attract small savers and borrowers away from traditional banks.

Government borrowing crowding out private credit. When the government borrows heavily from banks (for example through treasury securities), banks prefer this safe lending and less credit remains for private businesses — the **crowding-out** effect.

Weak collateral and land-title systems. Banks lend against guarantees, especially land and buildings. Because many properties lack registered land titles, potential borrowers cannot offer acceptable collateral, and banks cannot easily seize assets in case of default.

Consequences for the economy. These problems mean that credit creation in Cameroon is far

below its theoretical potential: investment by small and medium-sized enterprises is starved of finance, interest rates remain high, economic growth and job creation are slowed, and monetary policy transmitted through banks is weakened.

Possible remedies. Among the solutions often proposed are: speeding up land-title registration to widen access to collateral; creating and strengthening credit bureaux to share information on borrowers; encouraging the formalisation of small businesses; developing judicial procedures for faster loan recovery; promoting financial education to raise the banking rate; and closer cooperation between banks, microfinance institutions and mobile money operators to reach the unbanked population.

KEY POINTS **Key points of Section 2**

- Credit creation: banks create new deposits by lending their excess reserves.
- A single bank lends only its excess reserves; the system multiplies deposits.
- Credit multiplier: $M = \frac{1}{r}$; total deposits = initial deposit $\times M$; new credit = total deposits − initial deposit.
- The multiplier is reduced by cash leakages, excess reserves, weak loan demand and central bank regulation.
- Cameroonian banks face defaults, informality, low bancarisation, liquidity and cost pressures, competition from microfinance and mobile money, crowding out, and weak collateral systems — all of which limit real credit creation.

EXERCISE 2 **Credit creation and the multiplier**

A new deposit of 2 000 000 FCFA is made at a bank in Bamenda. The reserve ratio is 20%.

1. Calculate the credit multiplier.
2. Calculate the total deposits the banking system can support and the total credit newly created.
3. Present the T-accounts of the first two banks in the chain.
4. Explain two reasons why, in the real Cameroonian economy, the credit actually created would be smaller than your answer to question 2.

SOLUTION **Exercise 1**

1. $M = \frac{1}{0{,}2} = 5$.
2. Total deposits = 2 000 000 × 5 = 10 000 000 FCFA; new credit = 10 000 000 −

2 000 000 = 8 000 000 FCFA.
3. Bank A: deposit 2 000 000, reserves 400 000, loan 1 600 000. Bank B: deposit 1 600 000, reserves 320 000, loan 1 280 000.
4. Any two of: cash held by the public (leakages); banks keeping excess reserves; lack of creditworthy borrowers; central bank restrictions. Each reduces the amount redeposited or relent, so the real multiplier is below 5.

3 SECTION 3: THE CENTRAL BANK AND THE CASE OF BEAC (TU 32 — LESSON 92; TU 33 — GUIDED WORK 5)

In Section 2, we saw that commercial banks in Cameroon are powerful institutions: by granting credit, they create money, and this power can be a source of growth but also of instability — excessive credit creation can fuel inflation, while poorly managed banks can fail and destroy depositors' savings. A natural question then arises: who watches over the watchers? Who controls the total quantity of money in the economy, who rescues a bank in difficulty, and who guarantees that the banknote in your pocket in Douala is worth exactly the same in N'Djamena or Libreville? The answer is a single, very special institution: the **central bank**. In our part of the world, that institution is the **BEAC** — the Bank of Central African States — whose headquarters stands in Yaoundé.

3.1 Definition and Distinctive Features of a Central Bank

DEFINITION **Central bank**

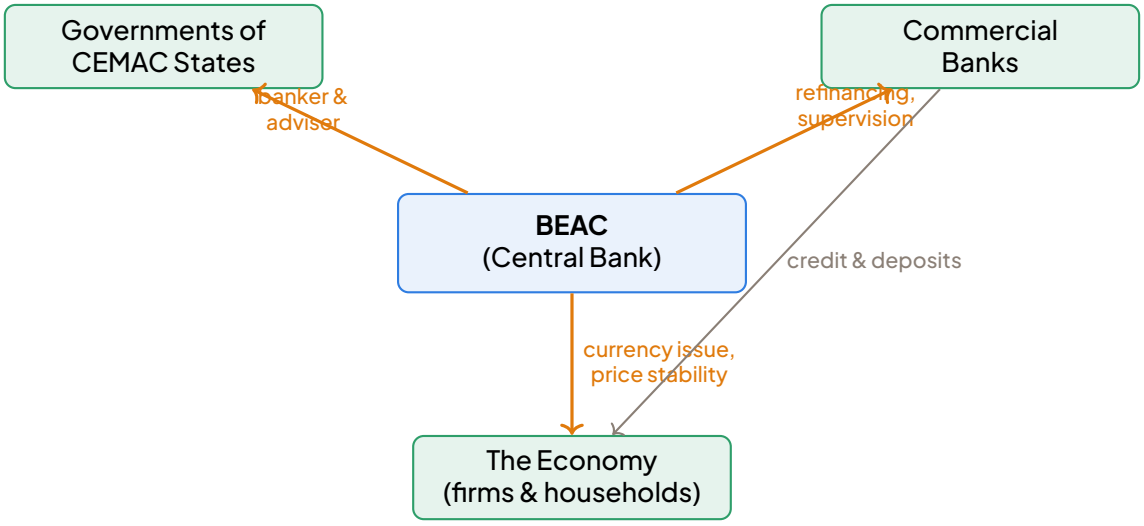
A **central bank** is the apex monetary institution responsible for issuing currency and for controlling the money supply and credit in an economy. It is a public (or publicly mandated) institution that does not seek profit and does not deal directly with ordinary individuals or firms.

To understand what makes a central bank different from a commercial bank, think of a football league. Commercial banks (Afriland First Bank, Société Générale Cameroun, BICEC…) are the players: they compete for deposits and customers and try to make profit. The central bank is the referee and rule-maker: it does not play to win, but without it the game would collapse into disorder. Three distinctive features summarise this special status:

- **Bank of issue:** it alone has the legal right to issue (print and mint) the currency that circulates in the economy. Every FCFA banknote you hold is a liability of the BEAC.
- **Bankers' bank:** commercial banks keep accounts at the central bank, borrow from it when short of liquidity, and settle payments between themselves through it.
- **Government's bank:** it keeps the accounts of the State, manages public debt operations and advises the government on financial matters.

WARNING **A frequent confusion**

The central bank is not a commercial bank and does not open accounts for individuals or businesses. If a candidate writes that "people deposit their savings at the BEAC", the examiner will penalise it. Ordinary customers deal with commercial banks; commercial banks deal with the central bank.



The central position of the BEAC: it links the government, the commercial banks and the economy.

3.2 Functions of a Central Bank

The features above translate into six great functions, each of which can be illustrated with the BEAC.

1. Monopoly of currency issue. The central bank alone prints banknotes and mints coins, and it guarantees their value. This monopoly avoids the chaos that would result if every bank issued its own notes. In CEMAC, only the BEAC issues the **FCFA** (franc of the Financial Cooperation in Central Africa). The BEAC determines the volume of fiduciary money in circulation based on the needs of the economy and its monetary policy stance.

2. Banker to commercial banks — lender of last resort. Commercial banks hold accounts at the central bank, through which interbank payments are settled, and they can borrow from it when they face a temporary shortage of cash.

DEFINITION **Lender of last resort**

The **lender of last resort** is the central bank's role of lending to commercial banks facing temporary liquidity crises, so that a solvent bank does not collapse simply because it can-

not meet sudden withdrawals. This function protects depositors and prevents a single bank's difficulties from spreading panic through the whole system.

At the BEAC, this role is fulfilled through the **refinancing facilities** (appels d'offres, guichets de prêt marginal, pensions livrées) where banks obtain central bank money against eligible collateral (government securities, eligible private bills).

3. Banker and adviser to the government. The central bank keeps the Treasury's accounts, manages the State's payments and public debt, and advises the government on economic and financial policy. The BEAC performs this role for each of the six CEMAC Treasuries. It manages the compte d'opérations of each Treasury and ensures the execution of budgetary expenditures.

4. Management of foreign exchange reserves. The central bank holds and manages the country's stock of foreign currencies (dollars, euros…) needed to pay for imports and to defend the external value of the currency. CEMAC's foreign reserves are pooled at the BEAC — a key feature of the monetary union. The BEAC invests these reserves in safe, liquid assets abroad. The level of gross foreign reserves (in months of imports) is a critical indicator of the zone's external sustainability.

5. Supervision of the banking system. The central bank licenses banks, sets prudential rules (minimum capital, liquidity ratios, risk concentration limits) and inspects them. In CEMAC this task is carried out by the **COBAC** (Commission Bancaire de l'Afrique Centrale — Central African Banking Commission), the supervisory arm attached to the BEAC. COBAC ensures compliance with the Règlement COBAC (prudential regulations) and can sanction or withdraw licences.

6. Conduct of monetary policy. This is the most macroeconomic function: the central bank regulates the quantity of money and credit in order to reach the objectives of monetary policy (see below). The BEAC's Monetary Policy Committee (CPM — Comité de Politique Monétaire) meets regularly to set the stance.

DID YOU KNOW? **GCE exam tip**

Whenever a question asks you to "explain the functions of a central bank", always illustrate each function with a BEAC example (issue of the FCFA, TIAO, COBAC, pooled reserves, SYSTAC payment system…). Application marks are reserved for candidates who root their answers in the Cameroonian/CEMAC context.

3.3 Objectives and Instruments of Monetary Policy

Objectives. Monetary policy pursues three great objectives, ranked by the BEAC's statutes:

- **Price stability**: keeping inflation low and predictable — the **primary objective** of the BEAC (inflation target around 3% per year for the zone);
- **Support to growth and employment**: ensuring that credit conditions allow firms to invest and create jobs, without prejudice to price stability;
- **External stability of the currency**: maintaining the value and convertibility of the FCFA, which

is pegged to the euro at a fixed parity (1 EUR = 655.957 FCFA).

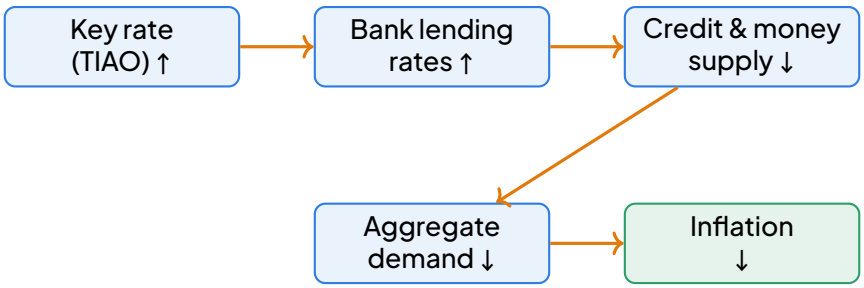
Instruments. To reach these objectives, the central bank disposes of five main instruments. At the BEAC, they operate within a liquidity management framework where the central bank steers short-term market rates:

- **The key (policy) interest rate — TIAO:** the rate at which the central bank lends to commercial banks through its main refinancing operations (appels d'offres hebdomadaires). At the BEAC it is called the **TIAO** (taux d'intérêt des appels d'offres — tender rate). When the TIAO rises, refinancing becomes dearer for banks, which raise their own lending rates.
- **The marginal lending rate — TIP:** the **TIP** (taux d'intérêt pénal) is the rate at which banks can obtain overnight liquidity from the BEAC on their own initiative, against eligible collateral. It acts as a ceiling for the interbank overnight rate.
- **The deposit facility rate — TIR:** the **TIR** (taux d'intérêt de référence) is the rate paid on banks' overnight deposits at the BEAC. It acts as a floor for the interbank overnight rate. The corridor (TIP – TIR) is typically 200 basis points.
- **Reserve requirements:** the obligatory fraction of deposits that each commercial bank must keep, unlent, at the central bank. In CEMAC, the **coefficient de réserves obligatoires** is currently 7% (subject to change by CPM decision). Raising the reserve ratio directly reduces the credit multiplier studied in Section 2. Reserves are remunerated at a rate linked to the TIAO.
- **Open market operations:** buying or selling securities (mainly Treasury bills and bonds) to inject or withdraw liquidity from the banking system. The BEAC conducts weekly appels d'offres (main refinancing operations) and can conduct pensions livrées (repos) or outright purchases/sales.
- **Refinancing / rediscount facilities:** the standing facilities through which banks obtain central bank money against eligible bills (TIP facility, rediscount of eligible commercial paper).
- **Moral suasion:** informal pressure, persuasion and directives addressed to banks to encourage desired behaviour (e.g. restraining credit to speculative activities, directing credit to priority sectors like agriculture or SMEs).

PROPERTY **Effects of monetary policy stances**

A **contractionary** (restrictive) monetary policy — higher key rate (TIAO), higher reserve requirements — reduces the money supply and credit, and is used to fight inflation. An **expansionary** policy — lower key rate (TIAO), lower reserve requirements — increases the money supply and credit, and is used to stimulate growth and employment. (The mechanism itself, not a formal proof, is examinable.)

The logic behind this property is the transmission mechanism of monetary policy, summarised below.



The transmission mechanism of a contractionary monetary policy.

METHOD **Analysing any central bank decision**

When the GCE paper presents a decision of a central bank (e.g. "the BEAC raises the TIAO"), proceed in three steps:

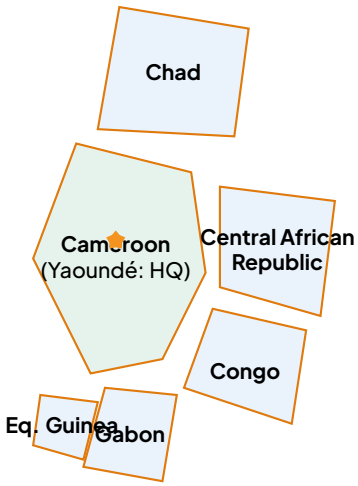
1. Identify the instrument used (key rate, reserve requirement, open market operation…);
2. Trace the direction of the effect on money supply and credit (expansion or contraction), using the transmission mechanism;
3. State the final objective targeted (fighting inflation, supporting growth, defending the currency…).

3.4 Guided Work 5: Case Study of the BEAC

Presentation of the institution. The **Bank of Central African States** (BEAC, Banque des États de l'Afrique Centrale) was created in **November 1972** as the successor of the Banque Centrale des États de l'Afrique Équatoriale et du Cameroun (BCEAEC). Its headquarters are in **Yaoundé, Cameroon**. It is the common central bank of the six member states of **CEMAC** (Communauté Économique et Monétaire de l'Afrique Centrale — Economic and Monetary Community of Central Africa): **Cameroon, Chad, the Central African Republic, Congo (Brazzaville), Gabon and Equatorial Guinea**. It issues the **FCFA**, which circulates identically in all six countries, and it cooperates with the **French Treasury**, which guarantees the convertibility of the FCFA into euros at a fixed parity — the anchor of the currency's external stability. The BEAC operates under the Statuts de la BEAC and the Convention de garantie with France.

WARNING **The classic mistake**

Do not write that the BEAC "belongs to Cameroon" or is "Cameroon's central bank". The BEAC is the common central bank of the six CEMAC states; its headquarters simply happen to be in Yaoundé. Confusing the seat of an institution with its ownership is a frequent error that costs marks.



Schematic map of the six CEMAC member states sharing the BEAC (star: Yaoundé, headquarters).

Governance. The BEAC is governed by a **Governor** (appointed by the Conference of Heads of State) and a **Board of Directors** (Conseil d'Administration). The **Monetary Policy Committee (CPM)** — composed of the Governor, Vice-Governors, and national directors — sets monetary policy. The **COBAC** (Commission Bancaire de l'Afrique Centrale) is the independent supervisory body.

BEAC in action. The BEAC's Monetary Policy Committee meets regularly and takes decisions that directly affect Cameroonian households and firms:

- it sets the **TIAO**, the key rate at which it refinances commercial banks through weekly tenders — in recent years (2022–2024) it has been raised several times (e.g., from 3.50% to 5.00%) to combat the inflationary pressures felt across CEMAC;
- it fixes the **TIP** (marginal lending rate) and **TIR** (deposit facility rate), defining the interest rate corridor;
- it fixes **reserve requirements** (coefficient de réserves obligatoires) for banks, thereby controlling their capacity to create credit;
- through **COBAC**, it supervises banks and microfinance institutions, sanctioning those that breach prudential rules (capital adequacy, liquidity, risk concentration);
- it manages the **pooled foreign exchange reserves** of the six states, essential for paying for imports of fuel, medicine and machinery;
- it operates the **SYSTAC** (Système de Télécompensation de l'Afrique Centrale), the real-time gross settlement (RTGS) system for large-value payments and the automated clearing house for retail payments.

WORKED EXAMPLE 3 **Analysing a BEAC decision**

Statement. Facing rising inflation in the CEMAC zone, the BEAC decides to increase its

key interest rate (TIAO) by 50 basis points. (a) Identify the instrument used. (b) Explain, step by step, how this decision is expected to reduce inflation. (c) State one possible cost of this policy for the Cameroonian economy.

SOLUTION

Solution. (a) The instrument is the key (policy) interest rate — the TIAO, the rate at which the central bank refinances commercial banks through its main refinancing operations (appels d'offres). (b) Transmission mechanism: the rise in the TIAO makes refinancing dearer for commercial banks; they react by raising their own lending rates (prime rate, overdraft rates); credit becomes more expensive, so households and firms borrow less; the money supply and aggregate demand grow more slowly; with less demand pressure on goods markets, inflation falls. (c) A possible cost: dearer credit discourages investment by Cameroonian firms (e.g., in agro-industry or manufacturing) and may slow economic growth and job creation in the short run — the classic trade-off between fighting inflation and supporting activity. If the inflation is cost-push (e.g., imported food/fuel prices), the policy may be less effective and more costly in terms of output loss.

Critical appreciation (exam enrichment). A strong GCE essay ends with evaluation. The BEAC presents clear strengths: a shared central bank gives six small economies a credible, stable currency; the French Treasury guarantee secures convertibility; pooling reserves protects each member better than national reserves would; the fixed peg eliminates exchange rate uncertainty within the zone and with the euro area. But there are also limits: one single monetary policy must suit six economies with different structures and shocks (oil-dependent Gabon or Equatorial Guinea versus more diversified Cameroon); the fixed peg to the euro removes the exchange rate as an adjustment tool (no depreciation to absorb external shocks); and decisions taken for the whole zone may not match Cameroon's specific needs at a given moment (asymmetric shocks). Mentioning this balance — without exaggeration — distinguishes an excellent script.

KEY POINTS **Key points of Section 3**

- A central bank is the apex monetary institution: bank of issue, bankers' bank, government's bank.
- Its six functions: currency issue, lender of last resort, banker/adviser to government, management of foreign reserves, banking supervision (via COBAC), conduct of monetary policy.
- Instruments: key rate (TIAO), marginal lending rate (TIP), deposit facility rate (TIR), reserve requirements (currently 7%), open market operations (appels d'offres, pensions livrées), refinancing facilities, moral suasion.
- Contractionary policy (higher TIAO, higher reserves) ⇒ less money/credit, fights inflation; expansionary policy (lower TIAO, lower reserves) ⇒ more money/credit, sup-

ports growth.

- The BEAC (1972, Yaoundé) is the common central bank of the six CEMAC states, issuing the FCFA under a convertibility guarantee of the French Treasury (fixed parity 1 EUR = 655.957 FCFA), with COBAC as supervisor and SYSTAC as payment system.
- Primary objective: price stability (inflation target 3%). Secondary: support growth, external stability.

EXERCISE 3 **Check your understanding**

1. Define a central bank and state its three distinctive features.
2. Explain the role of "lender of last resort" and why it protects depositors.
3. The BEAC lowers reserve requirements from 7% to 5%. Analyse the expected effects on credit creation, money supply and economic activity.
4. Give two strengths and two limits of sharing a common central bank within CEMAC.
5. Distinguish between the TIAO, TIP and TIR at the BEAC. What is the typical corridor width?

SOLUTION **Exercise n°3**

Lowering reserve requirements from 7% to 5% is an expansionary measure. Commercial banks are obliged to keep a smaller fraction of deposits unlent at the BEAC, so their excess reserves rise. Through the credit multiplier (Section 2), each franc of primary deposit now supports a larger volume of loans: the theoretical multiplier rises from $1/0.07 \approx 14.3$ to $1/0.05 = 20$. Credit creation expands and the money supply increases. Cheaper and more abundant credit encourages consumption and investment by Cameroonian households and firms, raising aggregate demand, output and employment. The risk, to be mentioned for full marks, is that if the economy is already near full capacity, the extra demand may translate into inflation rather than real growth.

SOLUTION **Exercise n°5**

The TIAO (taux d'intérêt des appels d'offres) is the main policy rate — the rate at which banks obtain liquidity through weekly tenders (main refinancing operations). The TIP (taux d'intérêt pénal) is the marginal lending facility rate — the rate for overnight borrowing on banks' own initiative, acting as a ceiling for the interbank overnight rate. The TIR

(taux d'intérêt de référence) is the deposit facility rate — the rate paid on banks' overnight deposits at the BEAC, acting as a floor. The typical corridor width is 200 basis points (TIP – TIR = 2%), with the TIAO usually set in the middle.

Having understood how the formal banking system is steered from the top by the BEAC, we turn in the next section to institutions that bring financial services to those whom commercial banks often neglect: the microfinance sector.

4 SECTION 4: MICROFINANCE ACTIVITIES IN CAMEROON (TU 34 — LESSON 93)

In the previous section, we studied the **central bank** and examined the case of the **BEAC**, the institution that issues currency, supervises commercial banks and conducts monetary policy in the CEMAC zone. Yet a striking paradox remains: despite the existence of a central bank and a network of commercial banks, the large majority of Cameroonians — small farmers in the North West, market women in Douala, motorcycle-taxi riders in Yaoundé, artisans in Bafoussam — have historically had no access to ordinary bank credit. Commercial banks demand collateral, payslips, business plans and minimum deposits that these economic agents simply cannot provide. It is precisely to fill this gap that **microfinance** emerged. This section defines microfinance, describes the categories of microfinance institutions (MFIs) operating in Cameroon, analyses their services and importance, and evaluates the problems they face.

4.1 Definition and Origin of Microfinance

DEFINITION **Microfinance**

Microfinance is the provision of small-scale financial services — savings, credit, insurance and money transfers — to low-income people who are excluded from the formal banking system. **Microcredit** is the specific component of microfinance that consists of granting very small loans to poor or low-income borrowers, usually to finance income-generating activities.

The intuition is simple. A commercial bank will not lend 50,000 FCFA to a woman selling roasted plantain by the roadside: the loan is too small to be profitable for the bank, and she has no land title or salary slip to offer as security. Yet that 50,000 FCFA may be exactly what she needs to buy a larger stock of plantain and charcoal, double her daily sales, and gradually lift her household out of poverty. Microfinance therefore rests on a powerful idea: the poor are bankable — they can save and they can repay, provided financial services are designed for their reality.

The modern microfinance movement is generally associated with the experience of the Grameen Bank founded by Muhammad Yunus in Bangladesh, which demonstrated that poor borrowers, especially women organised in small groups, repay loans at remarkably high rates when lending is based on **group guarantees** rather than physical collateral. The idea spread worldwide, and

in Cameroon it took root through cooperative movements, church-linked credit unions and village savings groups, building on the long tradition of tontines (rotating savings associations, known locally as njangi in the North West and South West).

PROPERTY **Social collateral in microcredit**

Microcredit typically relies on **group guarantees** and **social collateral** instead of physical collateral: borrowers form small solidarity groups whose members guarantee one another's loans, so that peer pressure and community reputation replace land titles and salary slips as security for repayment.

DID YOU KNOW? **The MC^2 model**

Cameroon developed its own celebrated microfinance model: the **MC^2** network (Mutuelles Communautaires de Croissance), born in the Bamiléké highlands of the West Region. In an MC^2, members of a community pool their savings in a community-owned fund and lend to one another for small projects. The model became so successful that it was studied and replicated in other African countries — a fine example of Cameroonian financial innovation.

4.2 Categories of Microfinance Institutions in Cameroon

Microfinance institutions in Cameroon take several organisational forms:

- **Credit unions and savings and credit cooperatives (SACCOs):** member-owned cooperatives that collect savings from members and grant them loans. The largest network is **CAMCCUL** (Cameroon Cooperative Credit Union League), which regroups hundreds of credit unions across the country.
- **Village banks and community funds:** small, community-based institutions (such as the MC^2) serving a single village or neighbourhood, often with strong support from NGOs or development partners.
- **NGOs and specialised microfinance companies:** non-governmental organisations and private companies that provide microcredit and related services, sometimes without collecting deposits.

DEFINITION **Credit union**

A **credit union** is a member-owned cooperative financial institution that pools the savings of its members in order to grant them loans at moderate interest rates. Members are simultaneously owners and customers: they elect the management committee and share in any surplus according to cooperative principles.

PROPERTY **The COBAC classification of MFIs**

In the CEMAC zone, MFIs are classified into three categories and supervised by the Ministry of Finance and the COBAC (Commission Bancaire de l'Afrique Centrale):

1. First category: cooperatives that both collect savings from members and grant them credit (e.g., credit unions affiliated to CAMCCUL).
2. Second category: institutions that grant credit only, financing their lending from their own funds or borrowing (they may not collect deposits from the public).
3. Third category: institutions that collect savings only, without granting credit.

WARNING **Common mistake: "MFIs are unregulated"**

A frequent error in GCE essays is to claim that microfinance institutions operate outside any legal framework. This is false. In Cameroon, MFIs are classified into the three COBAC categories above, must be licensed, and are supervised by the Ministry of Finance and the COBAC. What is true — and worth writing — is that some unlicensed or fraudulent institutions operate illegally alongside the regulated sector, which is a problem of enforcement, not an absence of regulation.

4.3 Services Offered by Microfinance Institutions

Modern MFIs in Cameroon offer a growing range of services tailored to low-income clients:

- **Micro-savings:** small, flexible deposit accounts with very low minimum balances, allowing market traders and farmers to save daily or weekly surpluses safely.
- **Microcredit:** small loans (from a few thousand to a few hundred thousand FCFA) for petty trade, agriculture, artisanal production or school fees, often disbursed through solidarity groups.
- **Micro-insurance:** cheap insurance products covering illness, death or crop failure, protecting poor households from shocks that would otherwise push them deeper into poverty.
- **Money transfers and mobile banking linkages:** many MFIs now partner with mobile money operators (MTN Mobile Money, Orange Money) so that members can deposit, withdraw and repay loans by telephone, extending services deep into rural areas.

4.4 Importance of Microfinance in Cameroon

Microfinance plays a strategic role in the Cameroonian economy:

1. **Financial inclusion:** MFIs reach rural populations, women and informal-sector operators whom commercial banks consider too risky or too costly to serve. Women, who dominate petty trade and food-crop farming, form the majority of members of many credit unions

and njangi groups.

2. **Financing of small businesses and agriculture:** microcredit finances market stalls, poultry projects, cocoa and food-crop farms, tailoring workshops and transport activities — the backbone of the informal economy that employs most Cameroonians.
3. **Job creation and poverty reduction:** by financing income-generating activities, microfinance helps households create self-employment, smooth consumption between harvests, pay school fees and accumulate small assets.
4. **Mobilisation of domestic savings:** MFIs transform scattered rural savings into loanable funds, deepening financial intermediation in areas where no commercial bank branch exists.

EXEMPLE **A microcredit chain in Bafoussam**

Mama Rose, a trader in Bafoussam main market, joins a solidarity group of five women at a local credit union. She borrows 100,000 FCFA for six months at a simple monthly interest rate of 2%, guaranteed by her group. The interest due is $100{,}000 \times 0.02 \times 6 = 12{,}000$ FCFA, so she repays 112,000 FCFA in total. She buys a stock of second-hand clothing, resells it over the season, earns a trading profit of about 60,000 FCFA, repays the loan with interest, and keeps roughly $60{,}000 - 12{,}000 = 48{,}000$ FCFA as net gain. On the strength of her repayment record, she qualifies for a larger loan of 200,000 FCFA the following season. This graduated lending — small first loan, larger loans after successful repayment — is the classic microcredit mechanism.

4.5 Problems of Microfinance Institutions in Cameroon

Despite their achievements, MFIs in Cameroon face serious difficulties:

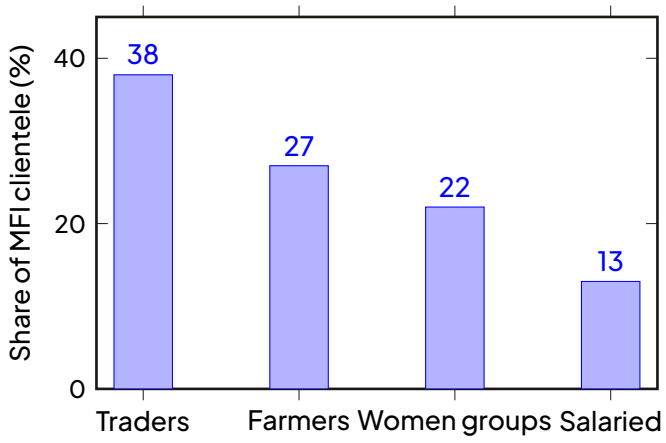
- **Weak management and governance:** many cooperatives are run by elected committees with limited training in accounting, credit analysis and risk management, leading to poor decisions and sometimes mismanagement of members' funds.
- **High default rates:** borrowers may divert loans to consumption, suffer business failure, or simply refuse to repay, especially where group solidarity weakens; defaults erode the capital of small MFIs.
- **Insufficient capital:** MFIs depend on members' savings and occasional donor support; their small capital base limits the size and number of loans they can grant and makes them fragile in a crisis.
- **Fraud and failures of unlicensed institutions:** fraudulent "wonder banks" and unlicensed operators have collected deposits and disappeared, destroying public confidence — the collapse of dubious savings schemes in the past made many Cameroonians wary of all MFIs.
- **High interest rates:** because small loans are costly to administer and default risk is high, MFI interest rates are often much higher than commercial bank rates, which can burden borrowers and limit the poverty-reduction impact.

4.6 Commercial Banks versus Microfinance Institutions

Commercial banks	Microfinance institutions
• Large firms, salaried workers • Large loans (millions of FCFA) • Physical collateral required • Lower interest rates • Supervised by COBAC / BEAC	• Low-income, informal sector • Small loans (thousands of FCFA) • Group / social collateral • Higher interest rates • Min. of Finance / COBAC categories

Comparison of commercial banks and microfinance institutions across key criteria

The two types of institution are complementary rather than rivals: commercial banks finance the formal sector and large projects, while MFIs bank the unbanked at the base of the economy. A successful micro-entrepreneur may eventually "graduate" from a credit union to a commercial bank.



Illustrative distribution of MFI clientele in Cameroon (traders 38%, farmers 27%, women groups 22%, salaried workers 13%) — indicative shares for classroom discussion, not official statistics

WORKED EXAMPLE 4 **Analysing the role of microfinance**

Statement. A credit union in Bamenda has 2,000 members. Each member saves on average 25,000 FCFA per year. The credit union lends out 80% of its deposits as microloans at an average interest rate of 18% per year, and pays 4% per year on members' savings.

(a) Calculate the total annual savings mobilised. (b) Calculate the amount lent out as microloans. (c) Calculate the credit union's gross interest margin on the lent funds. (d) State two reasons why this margin must be higher than that of a commercial bank.

SOLUTION

Solution.

1. Total savings = 2,000 × 25,000 = 50,000,000 FCFA (50 million FCFA).
2. Microloans granted = 0.80 × 50,000,000 = 40,000,000 FCFA.
3. Interest received on loans = 0.18 × 40,000,000 = 7,200,000 FCFA. Interest paid on savings = 0.04×50,000,000 = 2,000,000 FCFA. Gross interest margin = 7,200,000 − 2,000,000 = 5,200,000 FCFA.
4. The margin must be higher because (i) administering many very small loans is costly per franc lent (staff visits, group meetings, monitoring), and (ii) default risk is high since loans are granted without physical collateral, so the credit union must build reserves against losses.

METHOD **GCE exam tip: essays on financial inclusion**

When answering an essay on microfinance or financial inclusion:

1. Define microfinance and microcredit precisely.
2. Name concrete Cameroonian institutions: **CAMCCUL**, the **MC²** model, njangi/tontine traditions, mobile money linkages.
3. Use the COBAC three-category classification to show knowledge of the regulatory framework.
4. Balance the answer: importance (inclusion, jobs, rural finance) against problems (governance, defaults, fraud, high rates).
5. Conclude with a judgement, e.g., the need for stronger supervision and capacity-building.

KEY POINTS **Key points of Section 4**

- Microfinance provides small-scale savings, credit, insurance and transfer services to low-income people excluded from formal banking; microcredit is its lending component.

- Microcredit substitutes **group guarantees** and social collateral for physical collateral.
- Cameroonian MFIs include credit unions (CAMCCUL), village banks and MC^2, SACCOs, NGOs and specialised companies.
- COBAC classifies MFIs into three categories: savings-and-credit cooperatives, credit-only institutions, and savings-only institutions, supervised by the Ministry of Finance and COBAC.
- Microfinance promotes financial inclusion, finances small businesses and agriculture, and creates jobs, but suffers from weak governance, defaults, insufficient capital, fraud and high interest rates.

EXERCISE 4 **Essay practice**

(a) Distinguish between microfinance and microcredit. (b) Describe the three COBAC categories of microfinance institutions in Cameroon. (c) Discuss four contributions of microfinance to the Cameroonian economy and four problems faced by MFIs, using named examples such as CAMCCUL and the MC^2 model.

SOLUTION **Exercise 1**

(a) Microfinance is the whole range of small-scale financial services (savings, credit, insurance, transfers) for low-income people; microcredit is only the small-loan component. (b) First category: cooperatives collecting savings and granting credit; second category: credit-only institutions; third category: savings-only institutions. (c) Contributions: financial inclusion of rural people, women and informal operators; financing of petty trade and agriculture; job creation and poverty reduction; mobilisation of rural savings. Problems: weak management and governance; high default rates; insufficient capital; fraud by unlicensed operators; high interest rates. A strong answer names CAMCCUL credit unions and the MC^2 community funds, and concludes with the need for better supervision and training.

5 SECTION 5: APPRAISAL OF A LOAN FILE BY A CREDIT UNION (TU 35 — GUIDED WORK 6) AND CHAPTER SYNTHESIS

In the previous section, we explored the vibrant world of **microfinance institutions (MFIs)** in Cameroon — Credit Unions (COOPECs), microfinance banks, and financial NGOs. We saw how they mobilise savings from low-income households and provide financial services to the un-

banked and underbanked. However, the heart of any financial intermediary — whether a giant commercial bank like Afriland First Bank or a village Credit Union in Bafut — is **lending**. Lending transforms idle savings into productive investment. But every loan carries a risk: the borrower may not repay. For a Credit Union, where the capital belongs to the members themselves (teachers, farmers, market women), a single bad loan can erode trust and threaten the institution's survival. This is why **loan appraisal** — the rigorous, systematic evaluation of a borrower's file — is not a bureaucratic formality but the cornerstone of sustainability. In this final section, we master the art and science of credit appraisal, apply it to a realistic Cameroonian case, and then synthesise the entire financial system.

5.1 Why Loan Appraisal Matters: Protecting Members' Savings

DEFINITION **Loan Appraisal**

Loan appraisal (or credit appraisal) is the systematic process by which a financial institution evaluates a loan application to determine the borrower's creditworthiness — their ability and willingness to repay the loan according to the agreed terms — before deciding to approve, reject, or modify the request.

In a **Credit Union (COOPEC)**, the stakes are uniquely high:

- **Ownership structure:** The capital comes from members' savings (shares and deposits). A default directly reduces the value of members' shares or delays their withdrawals.
- **Social mission:** MFIs serve clients who often lack formal collateral (land titles, salaried payslips). Appraisal must rely on character and cash-flow analysis rather than just asset backing.
- **Regulatory requirement:** COBAC (Central African Banking Commission) regulations require MFIs to have written credit policies and to classify loans by risk. Poor appraisal leads to high **Non-Performing Loans (NPLs)**, triggering supervisory sanctions.

KEY POINTS **The Three Pillars of Sound Appraisal**

1. Protecting savers: Ensures that the money entrusted by a market woman in Douala or a cocoa farmer in Muyuka is lent only to viable projects.
2. Reducing default risk: Identifies red flags early (inflated sales figures, hidden debts, character issues).
3. Ensuring sustainability: Keeps the Portfolio at Risk (PAR > 30 days) low, allowing the MFI to cover its costs and pay dividends on shares.

5.2 The 5 Cs of Credit Appraisal: A Universal Framework

The **5 Cs** is the globally accepted framework for credit analysis. In the Cameroonian microfinance context, each "C" takes on a specific flavour.

METHOD **The 5 Cs Checklist for a Cameroonian Borrower**

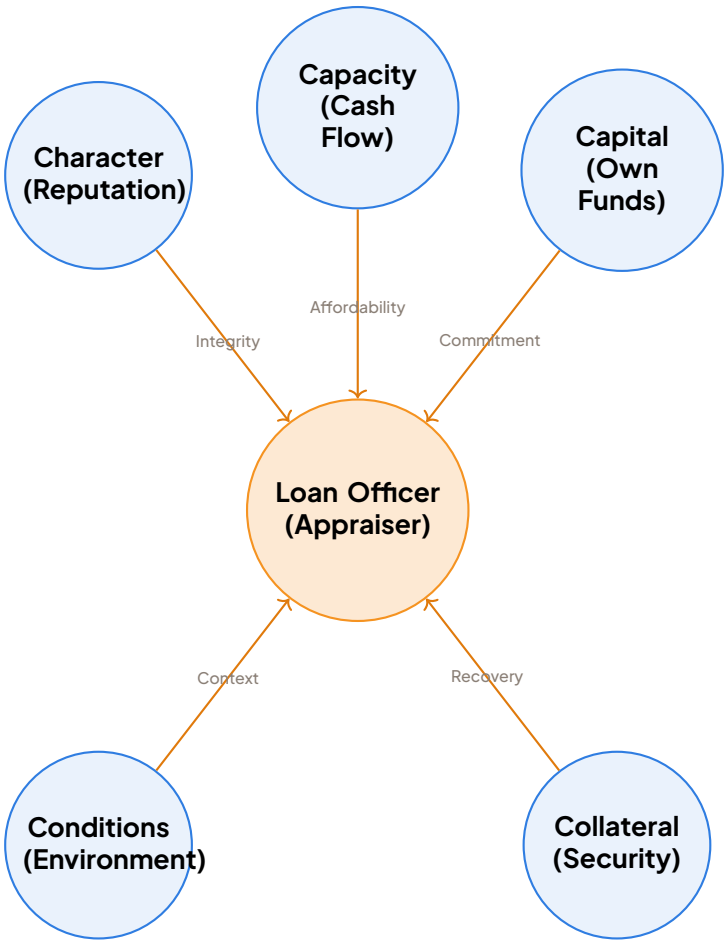
1. Character (Reputation & Willingness to Pay)
   Question: "Is this person honest and known to honour commitments?"
   Evidence: Membership duration, savings regularity, testimonials from group leaders (in solidarity groups), credit bureau check (if available via GIMAC or MFI internal database), community reputation. A member who has saved 5 000 FCFA weekly for 3 years without a single withdrawal shows stronger character than a newcomer demanding 500 000 FCFA.

2. Capacity (Repayment Ability)
   Question: "Does the business generate enough net cash flow to service the debt?"
   Analysis: Instalment-to-Income Ratio (IIR). Monthly instalment ≤ 30–35% of net monthly household/business income. For a farmer, capacity is seasonal — repayment schedule must align with harvest sales (e.g., cocoa main crop Oct–Jan).

3. Capital (Own Contribution / Skin in the Game)
   Question: "How much of their own money is the borrower risking?"
   Rule: Most MFIs require 10–30% own contribution (equity). If a trader wants 1 000 000 FCFA to buy a stock of frozen fish, they should contribute 200 000 FCFA from savings. This reduces moral hazard.

4. Collateral (Security / Last Resort)
   Question: "If the business fails, what can we recover?"
   Reality in Cameroon: Many MFI clients lack land titles (titre foncier). Acceptable collateral: group guarantee (solidarity), personal guarantors (salaried civil servants), chattel mortgage on equipment/vehicle (nantissement), blocked savings in the MFI (épargne bloquée). Warning: Collateral is a fallback, not the source of repayment.

5. Conditions (Economic & Policy Environment)
   Question: "What external factors could derail repayment?"
   Examples: Cocoa price crash on world market, fuel price hike increasing transport costs, insecurity in the Northwest/Southwest disrupting market access, new tax on mobile money affecting a POS agent's margins.



The 5 Cs of Credit Appraisal centred on the Loan Officer. Each "C" answers a distinct question; weakness in one cannot be fully compensated by strength in another.

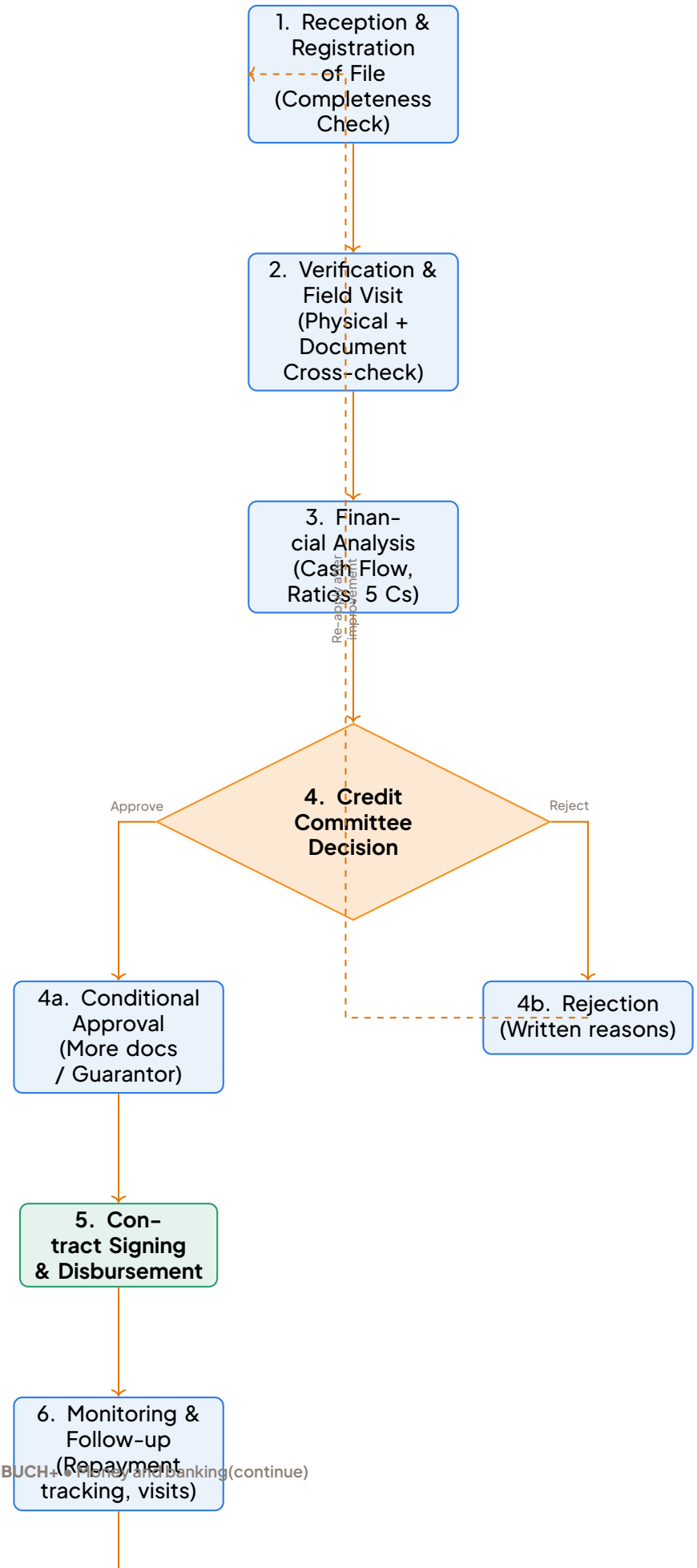
5.3 Elements of a Complete Loan File

A loan file (dossier de crédit) is the evidence base. Incomplete files are the number one cause of delays. The Credit Union's credit policy should specify the mandatory documents:

Document	Purpose / What the Appraiser Checks
1. Application Letter	Formal request: amount, purpose, duration, proposed repayment schedule. Signed by applicant.
2. Identification	Valid CNI (National ID) or passport; proof of residence (utility bill, quarter head attestation).
3. Membership & Savings History	Membership card, passbook or statement (min. 6–12 months). **Key:** Regularity, volume, voluntary vs. compulsory savings.
4. Business Plan / Purpose Statement	For business loans: description of activity, location, suppliers, customers, seasonality. For agriculture: crop calendar, acreage, input costs, expected yield.
5. Income & Cash-Flow Evidence	**Crucial.** Daily sales notebook (cahier de caisse), bank/MFI statements, purchase/sale invoices, mobile money (MoMo) transaction history. Appraiser reconstructs **Net Operating Income**.
6. Guarantors' Files	CNI, proof of income (payslip for civil servants), signed guarantee letter (lettre d'engagement de caution). Guarantor's own debt burden checked.
7. Collateral Documents	Land title (titre foncier) or certificat foncier, vehicle registration (carte grise), chattel mortgage deed (acte de nantissement), blocked savings agreement.
8. Field Visit Report	Loan officer's on-site notes: business exists? Stock matches claims? Premises condition? Neighbour testimony? GPS coordinates/photos.

5.4 The Appraisal Process: Step-by-Step Flowchart

The process must be standardised to avoid arbitrariness and ensure audit trails.


1. Reception & Registration of File (Completeness Check)
2. Verification & Field Visit (Physical + Document Cross-check)
3. Finan-cial Analysis (Cash Flow, Ratios, 5 Cs)
Re-apply after improvement
4. Credit Committee Decision
Approve
Reject
4a. Conditional Approval (More docs / Guarantor)
4b. Rejection (Written reasons)
5. Con-tract Signing & Disbursement
6. Monitoring & Follow-up (Repayment tracking, visits)

5.5 Worked Example: Appraising a Sample Loan File (Guided Work 6)

EXPERIMENT **Case Study: Appraising Mme Ngo's Loan Application**

Context: COOPEC "Espoir" in Bafoussam. Mme Ngo Sylvie, 42, member for 4 years, runs a wholesale/retail frozen fish business at Marché B. She applies for 2 000 000 FCFA over 12 months to buy a second-hand freezer (800 000 FCFA) and increase stock (1 200 000 FCFA) ahead of the Christmas peak.
File Contents:

- Savings: 350 000 FCFA (voluntary), regular weekly deposits of 5 000–10 000 FCFA. No withdrawals in 18 months.
- Business Records: Daily cash book for last 6 months. Average monthly sales: 1 800 000 FCFA. Average monthly purchases: 1 350 000 FCFA. Rent: 50 000 FCFA/month. Transport/Market fees: 80 000 FCFA/month. Household expenses (est.): 200 000 FCFA/month.
- Collateral: Husband (civil servant, Grade B2) offers personal guarantee + blocks 500 000 FCFA of his own savings in the COOPEC. Freezer to be purchased serves as chattel mortgage.
- Guarantors: Two fellow market traders (members >5 years) + husband.
- Conditions: Christmas season (high demand). Risk: Power cuts (freezer spoilage), fuel price hike (transport cost).

Task: As the Loan Officer, perform the appraisal using the 5 Cs and the Instalment-to-Income Ratio (IIR). Assume a monthly interest rate of 1.5% on declining balance (effective 1.5%/month, 18% p.a.). Monthly instalment ≈ 185 000 FCFA. Write your recommendation: Approve / Approve with Conditions / Reject.

WORKED EXAMPLE 5 **Detailed Appraisal of Mme Ngo's File**

Step 1: Character
Strong. 4-year member, disciplined saver (350 000 FCFA built from small weekly deposits), no withdrawal discipline. Market reputation verified by field visit: neighbours confirm she is "sérieuse". No past loans (first application) — clean slate.
Step 2: Capacity (The Quantitative Core)

Average Monthly Sales	= 1 800 000
Less: Cost of Goods Sold (Purchases)	= 1 350 000
Gross Margin	= 450 000
Less: Operating Expenses (Rent + Transport)	= 130 000
Net Business Income	= 320 000
Less: Household Expenses	= 200 000
Net Disposable Income (NDI)	= **120 000 FCFA/month**

Proposed Monthly Instalment = 185 000 FCFA.
Instalment-to-Income Ratio (IIR) = $\frac{185\,000}{120\,000} \times 100$ = **154**%.
RED FLAG! The instalment exceeds her net disposable income. She cannot repay from current cash flow without cutting household consumption drastically or the business growing significantly.
Step 3: Capital (Own Contribution)
Total project cost = 2 000 000 FCFA. Loan requested = 2 000 000 FCFA (100%). Zero own contribution. Violates typical MFI policy (min 10–20%). She is asking the MFI to bear all the risk.
Step 4: Collateral
Adequate on paper. Husband's blocked savings (500 000 FCFA) + chattel mortgage on freezer (800 000 FCFA resale value) + two member guarantors. Covers 1 300 000 FCFA of 2 000 000 FCFA (65%). But recall: collateral is not repayment source. Seizing a freezer in Bafoussam market takes months and yields low recovery.
Step 5: Conditions
Favourable but risky. Christmas peak (Dec) boosts sales 2–3×. However, power cuts (ENEO) are frequent — freezer failure = total stock loss. No backup generator in plan.
DECISION: APPROVE WITH CONDITIONS (Restructure)
Justification: Character is excellent, collateral acceptable, but Capacity fails (IIR > 100%) and Capital contribution is zero. Granting as requested sets her up for certain default.
Proposed Restructuring:

1. Reduce loan to 1 200 000 FCFA (finances freezer + partial stock). She contributes 300 000 FCFA from her savings (15% own contribution) and uses 500 000 FCFA from husband's blocked savings as internal collateral (not spent).
2. New instalment ≈ 110 000 FCFA/month. IIR = 110 000/120 000 = **92**% — still high but manageable if Christmas sales materialise.
3. Seasonal Repayment Schedule: Interest-only (or very low principal) Jan–Oct; balloon principal payments Nov–Dec from Christmas peak revenue.
4. Condition: Purchase voltage stabiliser for freezer (cost 50 000 FCFA included in loan).
5. Mandatory monthly field visit by loan officer during first 6 months.

Credit Committee votes: 4 in favour (with conditions), 1 abstention. File approved.

WARNING **Common Appraisal Mistakes — GCE Traps**

- Mistake 1: "Collateral substitutes for Capacity." Wrong. A loan officer who approves because "the husband blocked savings" ignores that the business must generate cash. Seizing collateral destroys the member relationship and yields 30–50% of value.
- Mistake 2: "Ignoring seasonality." Wrong. A cocoa farmer with 0 FCFA income Apr–Sep cannot pay equal monthly instalments. The schedule must match the cash-flow cycle.
- Mistake 3: "Accepting inflated sales figures." Wrong. Always cross-check: stock turnover, supplier invoices, MoMo history, physical stock count. "Oral accounting" is not evidence.
- Mistake 4: "Skipping the field visit." Wrong. Desk appraisal alone misses: business closed, different activity, ghost employee, location risk.

5.6 Common Causes of Loan Default in Cameroonian Credit Unions & Prevention

Cause of Default	Mechanism	How Good Appraisal Prevents It
1. Over-indebtedness / Multiple borrowing	Borrower takes loans from 2–3 MFIs + informal lenders; total debt service > income.	**Check credit bureau (GIMAC), ask for declaration of other debts, verify bank/MoMo statements for hidden repayments.**
2. Diversion of funds	Loan for "stock" used for wedding, funeral, house repair, or lent to a friend.	**Direct disbursement to supplier (cheque/transfer), field visit to verify stock purchase, post-disbursement monitoring visit at 2 weeks.**
3. Business failure due to external shock	Price collapse (cocoa, coffee), insecurity, fuel hike, pandemic.	**Conditions analysis:** Stress-test cash flow at -20% revenue. **Insurance linkage** (crop/index insurance). Flexible restructuring clauses.
4. Poor character / Moral hazard	Willful refusal to pay despite ability; "strategic default".	**Character check:** Long membership, savings discipline, community reputation, guarantor quality. **Group lending** (peer pressure).
5. Weak appraisal / "Tick-box" compliance	Officer rushes file to meet targets; committee rubber-stamps.	**Institutional safeguards:** Independent credit committee, mandatory field visit sign-off, portfolio quality audits, officer incentives tied to repayment rate not volume.

5.7 Chapter Synthesis: The Cameroonian Financial System — A Unified Map

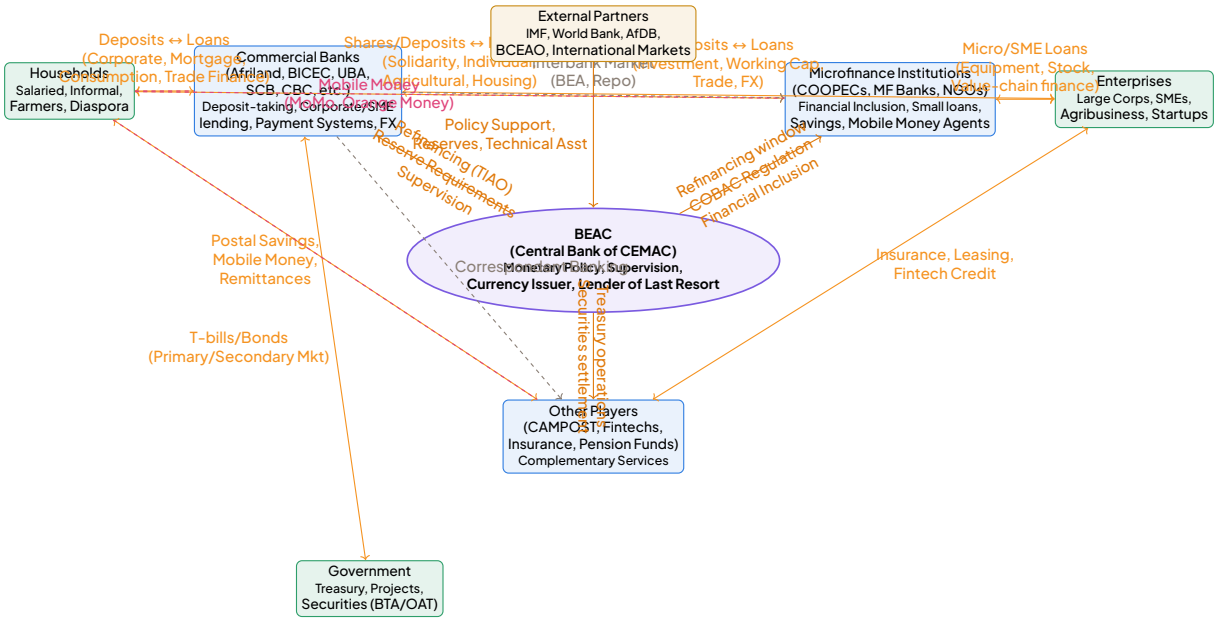
We have journeyed through the architecture of money and banking in Cameroon:

- **Section 1 (TU 29): Commercial Banks** — balance sheets, credit creation, the engine of broad

money (M_2 / M_3) expansion.

- **Section 2 (TU 30–31): Credit Creation Mechanics** — the money multiplier, reserve requirements, and the real-world frictions (excess reserves, leakages, NPLs) that limit it in Cameroon.
- **Section 3 (TU 32–33): The Central Bank (BEAC)** — the conductor: monetary policy (TIAO, reserve requirements, standing facilities), banking supervision (via COBAC), lender of last resort, guardian of the CFA franc peg.
- **Section 4 (TU 34): Microfinance (MFIs)** — the inclusive frontier: COOPECs, MF banks, reaching the last mile with solidarity lending, mobile money integration.
- **Section 5 (TU 35): Credit Appraisal** — the micro-foundation of asset quality; without it, the whole edifice crumbles.

These are not isolated silos. They form an **interconnected ecosystem**.



The Cameroonian Financial System: An Integrated Map. BEAC sits at the centre, steering monetary policy and supervising intermediaries. Commercial banks dominate corporate finance; MFIs drive inclusion. Households and firms access multiple channels. Mobile money (pink dashed) cuts across all layers.

KEY POINTS **Synthesis: Key Linkages for the GCE Exam**

1. Monetary Policy Transmission: BEAC raises TIAO → Commercial banks' refinancing cost ↑ → Lending rates ↑ → Credit to economy ↓ → Inflation ↓ (but also investment ↓). MFIs often lag in transmission due to fixed-rate methodologies.
2. Financial Inclusion Gap: Commercial banks require collateral/history → exclude 60%+ adults. MFIs fill gap with character-based lending. Mobile money is the new

bridge (digital savings/credit scoring).

3. Systemic Risk: A crisis in the cocoa sector → Farmer defaults at MFI → MFI liquidity crunch → MFI draws on BEAC refinancing or commercial bank line → Contagion risk. COBAC supervision of both banks and MFIs is the firewall.
4. Money Creation: Only commercial banks create scriptural money (demand deposits) via lending. MFIs create credit but their liabilities (shares/deposits) are not fully transactional money (no cheque book for COOPEC shares). BEAC creates base money (fiduciary + reserves).
5. Exam Essay Structure: If asked "Describe the Cameroonian financial system", structure: 1. Apex (BEAC/COBAC) → 2. Deposit-taking Intermediaries (Banks vs MFIs) → 3. Non-bank Intermediaries → 4. Markets (Money, Bond, FX) → 5. Payment Systems → 6. Challenges/Reforms (Inclusion, Digital, AfCFTA).

EXERCISE 5 **GCE-Style Practice Questions**

1. Case Study: Mr. Atanga, a cocoa farmer in Mamfe, applies for 500 000 FCFA at his COOPEC for fertiliser. He has 50 000 FCFA savings, no land title, but a salaried brother as guarantor. The loan officer approves without a farm visit. The cocoa price crashes 30%. Mr. Atanga defaults. Identify three appraisal failures and propose two policy changes for the COOPEC.
2. Essay: "The Central Bank is the pivot of the financial system." Discuss this statement with reference to BEAC's roles in monetary policy, banking supervision, and financial stability in the CEMAC zone.
3. Quantitative: A Credit Union has total loans of 500 million FCFA. Loans overdue > 30 days = 45 million FCFA. Loans overdue > 90 days = 20 million FCFA. Calculate PAR 30 and PAR 90. If the provisioning policy is 25% for 30–89 days and 100% for 90+ days, calculate the loan loss provision. Comment on the health of the portfolio.
4. Short Answer: Distinguish between character-based lending (microfinance) and collateral-based lending (traditional banking). Give one advantage and one limitation of each in the Cameroonian context.

SOLUTION **Exercise 3: PAR & Provisioning Calculation**

Data: Gross Loan Portfolio (GLP) = 500 million FCFA.
PAR 30 = 45 million. PAR 90 = 20 million.

Calculations:

$$\text{PAR 30 Ratio} = \frac{45}{500} \times 100 = \mathbf{9\%}$$

$$\text{PAR 90 Ratio} = \frac{20}{500} \times 100 = \mathbf{4\%}$$

Provisioning:

- Loans 30–89 days overdue = PAR 30 - PAR 90 = 45 - 20 = 25 million.
- Provision on 30–89 days = 25% × 25 = 6.25 million.
- Provision on 90+ days = 100% × 20 = 20 million.
- Total Loan Loss Provision = 6.25 + 20 = 26.25 million FCFA.

Comment: PAR 30 at 9% is above the COBAC prudential threshold (typically ≤ 5–10% depending on MFI category; generally > 10% is critical). PAR 90 at 4% signals a significant stock of chronic bad debts. The provision coverage (26.25 / 45 = 58%) is partial. The MFI must intensify recovery on the 90+ bucket and review its appraisal standards to stem new NPL formation.

DID YOU KNOW? **Did You Know? The "Caisse d'Épargne" Legacy**

Cameroon's first savings institution was the Caisse d'Épargne du Cameroun, created in 1928 under French mandate. It was the ancestor of today's CAMEC (Caisse d'Épargne et de Crédit) and the postal savings network (CAMPOST). The modern COOPEC movement took off in the 1960s–70s inspired by the Caisses Populaires of Quebec (Desjardins model) and German Genossenschaftsbanken (Raiffeisen). The first COOPEC, COOPEC "La Réussite", was founded in 1963 in Njinikom (Northwest) by Dutch missionaries. Today, the Réseau des Caisses Populaires du Cameroun (RECAP) and CamCCUL are among Africa's largest MFI networks, serving over 2 million members — a testament to the power of member-owned, character-based finance.

KEY POINTS **GCE Advanced Level: Final Exam Strategy for This Chapter**

- Definitions (2–3 marks): Learn verbatim: Money, Commercial Bank Balance Sheet items, Money Multiplier, TIAO, Required Reserves, Credit Union, Loan Appraisal, 5 Cs, PAR 30/90.
- Mechanisms (5–7 marks): Credit Creation step-by-step (T-account style). Monetary Policy Transmission chain (TIAO → Rates → Credit → AD → P/Y). BEAC Tools (Open Market Ops, Standing Facilities, Reserve Req).
- Calculations (4–6 marks): Money Multiplier ($1/rr$), Excess Reserves, PAR Ratios, Pro-

visioning, Instalment-to-Income Ratio, Effective Interest Rate (flat vs declining).

- Essay/Case Study (10–15 marks): Structure: Introduction (Definitions + Context) → Body: 3–4 argued points with Cameroonian examples (Cocoa, BEAC, COOPEC, MoMo) → Evaluation (Limits, Challenges, Policy Recommendations) → Conclusion.
- Keywords to sprinkle: **CEMAC**, **COBAC**, **CFA Franc**, **Fixed Exchange Rate**, **Financial Inclusion**, **Non-Performing Loans**, **Mobile Money**, **AfCFTA**, **Monetary Policy Committee (MPC)**.

End of Chapter ECO04: Money and Banking (Continue). You have now covered the full syllabus from commercial bank balance sheets to the appraisal desk of a village Credit Union, with the Central Bank orchestrating in between. Master the diagrams, practise the calculations, and always ground your answers in the **Cameroonian reality**. Good luck for the GCE!

6 INTEGRATION ACTIVITY

6.1 Aims of the activity

This integration activity brings together in a single, coherent case study all the notions studied in this chapter: the **balance sheet** of a commercial bank, the mechanism of **credit creation** and the **credit multiplier**, the role of the **central bank** and the **transmission mechanism** of monetary policy, the place of **microfinance** in Cameroon, and the **appraisal of a loan file** using the 5 Cs method.

By the end of the activity, you should be able to:

- construct and interpret the simplified T-account of a commercial bank after a new deposit, distinguishing **required reserves** from **excess reserves**;
- calculate the maximum credit creation of the banking system from a given reserve requirement, using the credit multiplier $k = \frac{1}{r}$;
- explain, step by step, how an increase in the key rate of the BEAC is transmitted to households, firms and the general price level in Cameroon;
- appraise a real loan file at a credit union using the 5 Cs (Character, Capacity, Capital, Collateral, Conditions) and defend an accept/reject decision with arguments;
- produce a short, well-structured written report combining calculation, economic analysis and a justified decision — exactly the skill required in the GCE Advanced Level essay and data-response papers.

KEY POINTS **Key formulae used in this activity**

- Required reserves = reserve ratio × deposits, i.e. $R = r \times D$.
- Excess (loanable) reserves = total new reserves − required reserves.
- Credit multiplier: $k = \frac{1}{r}$, where r is the reserve requirement expressed as a decimal.
- Maximum total deposits = k × initial new deposit; maximum new credit = maximum total deposits − initial deposit.
- Flat-rate monthly instalment = $\frac{\text{principal}}{\text{number of months}}$ + (principal × monthly flat rate).

6.2 Situation

From Mama Put's Deposit to a Taxi Loan.

Madame Eyenga runs a successful mama put (cooked-food) restaurant near the Mokolo market in Yaoundé. After several good months, she has accumulated **500,000 FCFA** in cash, which

she keeps at home. Worried about theft, she deposits the whole amount on 3 January in a current account at the Mokolo branch of **Afriland First Bank**.

The Bank of Central African States (**BEAC**), the central bank common to the six CEMAC countries (Cameroon, Chad, Central African Republic, Congo, Equatorial Guinea and Gabon), imposes on commercial banks a **reserve requirement of 13 %**: every bank must keep 13 % of its deposits as reserves (at the central bank or in vault cash) and may lend out the rest. Assume that:

- banks lend out all their excess reserves (no voluntary idle reserves);
- the public holds no cash: every loan is spent and re-deposited in full in the banking system;
- the banking system is initially in equilibrium.

Three months later, worried about inflationary pressures in the CEMAC zone (imported food and fuel prices are rising), the **Monetary Policy Committee of the BEAC raises its key policy rate (the TIAO, the interest rate on its main refinancing operations) by 50 basis points**, from 5.00 % to 5.50 %. Recall that **one basis point** is one hundredth of a percentage point: 50 basis points = 0.50 percentage point.

Meanwhile, Ndifor, a 27-year-old motorcycle-taxi (bendskin) rider in Bamenda, wants to buy a second motorcycle costing **800,000 FCFA** to rent to another rider. He applied to a commercial bank but was refused: he has no payslip, no land title and no formal accounts. He therefore applies for an **800,000 FCFA loan over 24 months at 1.5 % per month (flat) at a CAMCCUL-affiliated credit union**. His loan file contains:

Item	Details
Occupation	Motorcycle-taxi rider, 4 years' experience, known at his stage
Daily net income	About 6,000 FCFA/day, worked 26 days/month (≈ 156,000 FCFA/month)
Monthly expenses	Food, rent, fuel, maintenance: ≈ 100,000 FCFA
Savings history	Member of the credit union for 3 years; regular savings of 120,000 FCFA; one previous loan of 300,000 FCFA fully repaid on time
Guarantors	Two members of the union, each with savings above 200,000 FCFA
Collateral	No land title; offers the motorcycle itself (logbook deposited)
Business plan	Rent the new bike at 2,000 FCFA/day; expected ≈ 52,000 FCFA/month extra income
Monthly instalment	Principal + interest ≈ 45,333 FCFA/month for 24 months

Each group of 3–4 students acts successively as (i) the bank's accounts officer, (ii) an economist at the BEAC, and (iii) the credit committee of the credit union. Each group submits a **one-page report** combining the calculation, the policy analysis and the appraisal decision.

6.3 Tasks

1. **The bank's T-account.** Record Madame Eyenga's deposit of 500,000 FCFA in the simplified balance sheet (T-account) of Afriland First Bank, distinguishing required reserves from

excess reserves. How much can the bank lend immediately?

2. **Credit creation.** Assuming the loan is spent and re-deposited in the banking system, calculate the credit multiplier $k = \frac{1}{r}$ with $r = 13\,\%$. Deduce (a) the total amount of deposits the banking system can ultimately support, and (b) the total new credit created. Show the first three rounds of the process in a small table.

3. **Limits of the multiplier.** Give two reasons why, in the real Cameroonian economy, actual credit creation is smaller than the theoretical maximum.

4. **Monetary policy transmission.** The BEAC raises its key rate by 50 basis points. Explain the transmission mechanism step by step: effect on commercial banks' refinancing cost, on lending rates, on Madame Eyenga (saver), on borrowers like Ndifor, on investment and consumption, and finally on inflation in Cameroon. State one reason why this transmission may be weak in Cameroon.

5. **Loan appraisal.** Using the 5 Cs, appraise Ndifor's loan file. Compute his monthly repayment capacity (income minus expenses, before and after the new business income) and compare it with the monthly instalment of 45,333 FCFA.

6. **Decision.** As the credit committee, decide: accept, accept with conditions, or reject. Justify your decision in at most ten lines, proposing any guarantees or conditions you judge necessary.

7. **Synthesis.** In five lines, explain how this case study shows the complementarity between commercial banks, the central bank and microfinance institutions in financing the Cameroonian economy.

6.4 Model answer

1. The bank's T-account.

The deposit creates a liability (the current account owed to Madame Eyenga) and an asset (cash/reserves held by the bank). Required reserves are

$$R = r \times D = 0.13 \times 500{,}000 = 65{,}000 \text{ FCFA};$$

excess reserves are $500{,}000 - 65{,}000 = 435{,}000$ FCFA, which the bank can lend immediately.

Assets (FCFA)		Liabilities (FCFA)	
Required reserves	65,000	Demand deposit (Mme Eyenga)	500,000
Excess reserves / loanable	435,000		
Total	500,000	**Total**	500,000

Note that the balance sheet always balances: total assets = total liabilities = 500,000 FCFA.

2. Credit creation.

The credit multiplier is

$$k = \frac{1}{r} = \frac{1}{0.13} \approx 7.69.$$

(a) Total deposits the system can ultimately support:

$$500{,}000 \times 7.69 \approx 3{,}846{,}154 \text{ FCFA.}$$

(b) Total new credit created:

$$3{,}846{,}154 - 500{,}000 \approx 3{,}346{,}154 \text{ FCFA,}$$

equivalently $435{,}000 \times 7.69 \approx 3{,}346{,}154$ FCFA.

Round	New deposit	Required reserves	New loan
1 (Afriland)	500,000	65,000	435,000
2	435,000	56,550	378,450
3	378,450	49,199	329,252
··· Total	3,846,154	500,000	3,346,154

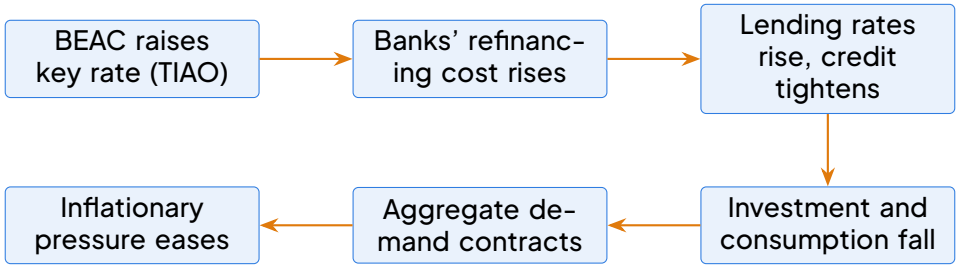
Each round, the new loan of one bank becomes the new deposit of the next; 13 % is immobilised as reserves and 87 % is re-lent. The process converges because each round is smaller than the previous one (it is a geometric progression with ratio $1 - r = 0.87$).

3. Limits of the multiplier.

(a) Cash leakages: in Cameroon many agents keep part of their money in cash (large informal sector, cash-based retail trade), so loans are not fully re-deposited and each round leaks out of the banking system. (b) Prudence of banks and lack of creditworthy borrowers: banks voluntarily hold excess reserves as a precaution, or cannot find solvent borrowers with acceptable guarantees — especially small informal businesses without land titles or payslips, like Ndifor. Both leakages reduce the effective multiplier well below 7.69.

4. Transmission of the 50-basis-point increase.

When the BEAC raises its key rate, commercial banks that refinance at the central bank pay more for the funds they borrow; their cost of funds rises, so they increase their lending rates and may tighten credit conditions (stricter files, more collateral demanded). For **Madame Eyenga** (a net saver), deposit rates may rise slightly — a gain. For **borrowers like Ndifor**, bank credit becomes dearer and harder to obtain, pushing them further towards microfinance. Firms face costlier investment credit, so investment falls; households reduce consumption financed by credit. Aggregate demand contracts, which eases demand-pull pressure on prices and helps bring **inflation** back towards the CEMAC convergence criterion (an inflation rate of at most 3 %).



The transmission mechanism of a restrictive monetary policy (interest-rate channel).

Weakness of transmission in Cameroon: a large share of economic activity is informal and financed outside the banking system (tontines, cash holdings, microfinance), and bank credit to the private sector remains limited, so a change in the key rate directly affects only the banked part of the economy. Moreover, when inflation is imported (fuel, food), a demand-side instrument has limited power over it.

5. Appraisal with the 5 Cs.

First, check the instalment: with a flat rate of 1.5 % per month on 800,000 FCFA over 24 months,

$$\text{monthly instalment} = \frac{800{,}000}{24} + (800{,}000 \times 0.015) = 33{,}333 + 12{,}000 = 45{,}333 \text{ FCFA},$$

which matches the figure in the file. Then appraise:

- **Character**: excellent — 3 years' membership, regular savings, previous loan of 300,000 FCFA repaid on time.
- **Capacity**: current disposable surplus ≈ 156,000 − 100,000 = 56,000 FCFA/month, which already covers the instalment of 45,333 FCFA (a margin of about 10,667 FCFA); with the rental income (≈ 52,000 FCFA/month), the surplus rises to about 108,000 FCFA, giving a comfortable safety margin even if the second bike earns nothing for some months.
- **Capital**: savings of 120,000 FCFA, i.e. 15 % of the loan — an acceptable personal contribution showing commitment.
- **Collateral**: no land title, but the motorcycle's logbook is deposited and two solvent guarantors co-sign — typical microfinance substitute guarantees.
- **Conditions**: the bendskin business is in strong demand in Bamenda; risks exist (accidents, breakdowns, insecurity), which justify insurance and maintenance provisions.

6. Decision.

Accept with conditions. The file shows good character, sufficient repayment capacity even without the new income, and acceptable guarantees. Conditions: (i) comprehensive insurance on the motorcycle; (ii) deposit of the logbook at the union; (iii) the two guarantors sign the bond; (iv) monthly instalments deducted first from his savings account; (v) disbursement directly to the motorcycle dealer to ensure the loan is used as planned.

7. Synthesis.

The case shows a complete financial circuit: the trader's deposit feeds a commercial bank, which multiplies credit under the reserve rules fixed by the BEAC; the central bank steers the cost and quantity of money through its key rate; and microfinance institutions like CAMCCUL credit unions reach clients excluded from banks, using savings history and solidarity guarantees instead of classical collateral. Together, these three levels channel savings towards investment and support growth in Cameroon.

WARNING **Common mistakes to avoid**

Do not confuse the first loan (435,000 FCFA) with the total credit created (≈ 3,346,154 FCFA). In the multiplier formula, use r as a decimal (0.13, not 13). Remember that 50 basis points means 0.50 percentage point, not 50 %. Finally, in the appraisal,

never decide on collateral alone: in microfinance, **character** and **capacity** come first.

EXERCISE 6 **For consolidation**

Suppose the BEAC lowers the reserve requirement from 13 % to 10 % and Madame Eyenga's deposit remains 500,000 FCFA, all other assumptions unchanged.

1. Recompute the credit multiplier, the maximum total deposits and the maximum new credit.
2. Is this measure expansionary or restrictive? Explain its likely effect on credit and economic activity in Cameroon.
3. A credit union lends 600,000 FCFA over 12 months at 2 % per month (flat). Compute the monthly instalment and the total interest paid.

SOLUTION **Exercise: For consolidation**

1. $k = \frac{1}{0.10} = 10$. Maximum total deposits = 500,000 × 10 = 5,000,000 FCFA; maximum new credit = 5,000,000 − 500,000 = 4,500,000 FCFA.
2. Expansionary: lowering the reserve requirement frees excess reserves, raises the multiplier and allows banks to grant more credit, which stimulates investment, consumption and growth (but may fuel inflation if demand outpaces supply).
3. Monthly instalment = $\frac{600{,}000}{12} + (600{,}000 \times 0.02) = 50{,}000 + 12{,}000 = 62{,}000$ FCFA. Total interest = 12,000 × 12 = 144,000 FCFA.

7 METHODS AND WORKED EXAMPLES

8 MONEY AND BANKING (CONTINUED)

8.1 The Balance Sheet of a Commercial Bank

DEFINITION **Commercial Bank Balance Sheet**

A **balance sheet** is a snapshot of a bank's financial position at a given date. It records **assets** (uses of funds) on the left and **liabilities** (sources of funds) on the right. The fundamental identity holds:

$$\text{Total Assets} \equiv \text{Total Liabilities} + \text{Shareholders' Equity}.$$

Assets are ordered by decreasing liquidity; liabilities by increasing urgency of repayment.

PROPERTY **Main Items in a Cameroonian Commercial Bank's Balance Sheet**

- Assets (Emplois): Cash in vault (valeurs en caisse); Reserves at BEAC (réserves obligatoires et excédentaires); Treasury bills and government securities (titres publics); Loans and overdrafts to customers (crédits à la clientèle); Bills discounted (effets escomptés); Fixed assets (immobilisations corporelles et incorporelles).
- Liabilities (Ressources): Sight deposits (dépôts à vue); Savings deposits (dépôts d'épargne); Time deposits (dépôts à terme); Borrowings from BEAC and other banks (emprunts auprès de la BEAC et banques); Share capital (capital social); Reserves and retained earnings (réserves et reports à nouveau); Provisions.

METHOD **Reading and Analysing a Bank Balance Sheet**

1. Draw the two-column framework: Assets (left) and Liabilities + Equity (right).
2. Classify every item: deposits are liabilities (the bank owes them); loans, securities, cash are assets (the bank owns claims).
3. Order assets from most liquid (cash, BEAC reserves) to least liquid (fixed assets, long-term loans). Order liabilities from most stable (equity, time deposits) to most re-

payable on demand (sight deposits).

4. Compute key prudential ratios:
   - **Liquidity ratio** (COBAC standard): $\frac{\text{Liquid assets (cash + BEAC reserves + Treasury bills)}}{\text{Short-term liabilities (sight + savings + time deposits)}} \geq 100\%$ (simplified).
   - **Solvency / Capital adequacy ratio** (Basel/COBAC): $\frac{\text{Regulatory capital (Tier 1 + Tier 2)}}{\text{Risk-weighted assets}} \geq 8\%$ (or 10% in CEMAC).
5. Interpret: high loans/assets → profitability but liquidity risk; high cash/assets → safety but low return.

EXEMPLE **T-Account Representation of a Deposit**

When a customer deposits 1 000 000 FCFA in cash at a bank:

Assets		Liabilities	
Cash in vault	+1 000 000	Sight deposits	+1 000 000

The bank's liquidity increases; its obligation to the customer increases by the same amount.

WORKED EXAMPLE 6 **Balance Sheet of Sanaga Commercial Bank**

The following items (in millions of FCFA) belong to Sanaga Commercial Bank, Douala: sight deposits 40 000; cash in vault 3 000; loans to customers 45 000; savings deposits 25 000; reserves at BEAC 7 000; shareholders' capital 10 000; treasury bills held 8 000; time deposits 15 000; buildings and equipment 5 000; reserves (retained profits) 12 000. Present the balance sheet in regulatory order and comment on liquidity and solvency.

SOLUTION

Step 1 — Classify and order.

- Assets (by liquidity): Cash in vault (3 000); Reserves at BEAC (7 000); Treasury bills (8 000); Loans to customers (45 000); Buildings & equipment (5 000).
- Liabilities + Equity (by stability): Share capital (10 000); Reserves/retained profits (12 000); Time deposits (15 000); Savings deposits (25 000); Sight deposits (40 000).

Step 2 — Present the balance sheet.

Assets (Emplois)	M FCFA	Liabilities + Equity (Ressources)	M FCFA
Cash in vault (Caisse)	3 000	Share capital (Capital social)	10 000
Reserves at BEAC (Réserves BEAC)	7 000	Reserves & retained profits (Réserves & RAN)	12 000
Treasury bills (Titres publics)	8 000	Time deposits (Dépôts à terme)	15 000
Loans to customers (Crédits clientèle)	45 000	Savings deposits (Dépôts d'épargne)	25 000
Buildings & equipment (Immobilisations)	5 000	Sight deposits (Dépôts à vue)	40 000
Total Assets	68 000	Total Liabilities + Equity	68 000

Step 3 — Compute ratios and interpret.

- Liquidity ratio (simplified): Liquid assets = 3 000 + 7 000 + 8 000 = 18 000. Short-term liabilities = 40 000 + 25 000 + 15 000 = 80 000. Ratio = 18 000 / 80 000 = 22.5%. Below the COBAC 100% norm — the bank relies on the interbank market or BEAC refinancing to meet withdrawals.
- Loans / Total assets = 45 000 / 68 000 ≈ 66.2%. High transformation ratio: profitable but vulnerable to a run.
- Equity / Total assets = (10 000 + 12 000) / 68 000 ≈ 32.4%. Strong capital base, well above the 8–10% risk-weighted requirement.

Conclusion: The bank is solvent but structurally illiquid — a typical profile for a Cameroonian commercial bank engaged in maturity transformation.

8.2 Credit Creation by Commercial Banks

DEFINITION **Credit Creation**

Credit creation is the process by which the banking system, starting from an initial deposit, grants loans that become new deposits in the system, thereby multiplying the money supply. It rests on **fractional reserve banking**: banks keep only a fraction r (reserve ratio) of deposits as reserves and lend out the rest.

PROPERTY **The Money (Credit) Multiplier**

Under the assumptions: (i) no cash drain (public holds no currency), (ii) banks hold zero excess reserves, (iii) all loans are re-deposited in the banking system, (iv) single reserve

ratio r for all deposits,

Money multiplier $k = \frac{1}{r}$, Total deposits created $D = k \times D_0 = \frac{D_0}{r}$, Total credit created $C = D - D_0 = D_0 \left($

With a **cash-drain ratio** c (fraction of loans the public keeps as cash) and/or **excess reserve ratio** e, the effective multiplier becomes

$$k_{\text{eff}} = \frac{1}{r + c + e}.$$

METHOD **Step-by-Step Credit Creation Calculation**

1. Identify the **initial new deposit** D_0 (or initial excess reserves) and the **required reserve ratio** r set by BEAC.
2. State the assumptions (no leakages, or specify c and e).
3. Compute the multiplier $k = 1/(r + c + e)$.
4. Compute total deposits $D = kD_0$ and total credit created $C = D - D_0$.
5. (Optional) Show the round-by-round process in a table to illustrate the geometric series.

EXPERIMENT **Simulating Credit Creation Round by Round**

Consider $D_0 = 500$ million FCFA, $r = 10\%$, no leakages.

Round	New Deposit	Reserves (10%)	New Loan	Cumulative Deposits
Initial	500	50	450	500
1	450	45	405	950
2	405	40.5	364.5	1355
3	364.5	36.45	328.05	1719.5
⋮	⋮	⋮	⋮	⋮
Limit	5 000	500	4 500	5 000

The geometric series $500(1 + 0.9 + 0.9^2 + \cdots) = 500/(1 - 0.9) = 5\,000$.

WORKED EXAMPLE 7 **Credit Creation in the Cameroonian Banking System**

An initial new deposit of 500 million FCFA is made at a commercial bank in Yaoundé. The required reserve ratio fixed by BEAC is 10%.

1. Calculate the money multiplier, total deposits and total credit created assuming no leakages.
2. Repeat with a cash-drain ratio of 5% (public keeps 5% of each loan in cash) and zero excess reserves.
3. Explain why the actual credit creation in Cameroon is often lower than the theoretical maximum.

SOLUTION

Part 1 — No leakages.

$k = \frac{1}{0.10} = 10$, $D = 10 \times 500 = 5\,000$ million FCFA, $C = 5\,000 - 500 = 4\,500$ million FCFA.

Part 2 — With cash drain $c = 0.05$.

$k' = \frac{1}{0.10 + 0.05} = \frac{1}{0.15} \approx 6.667$, $D' \approx 6.667 \times 500 = 3\,333$ million FCFA, $C' \approx 2\,833$ million FCFA.

Part 3 — Why actual creation is lower in Cameroon.

- High cash-drain ratio: a large informal sector means much currency circulates outside banks (c high).
- Excess reserves: banks often hold reserves above the minimum due to uncertainty, poor interbank market, or lack of creditworthy borrowers ($e > 0$).
- Non-bank financial intermediaries: microfinance institutions, tontines (njangui) capture savings that do not enter the commercial bank multiplier.
- BEAC prudential limits: concentration risk limits, sectoral credit ceilings, and the COBAC liquidity ratio constrain lending.
- Structural factors: weak collateral registries, high NPLs (non-performing loans), and information asymmetry ration credit.

WARNING **Common Mistakes on Credit Creation**

- Confusing total deposits created (D) with credit created ($C = D - D_0$).
- Forgetting that the multiplier applies to the banking system as a whole, not to a single bank (a single bank can only lend its excess reserves).

- Using the reserve ratio as a percentage (10) instead of a decimal (0.10) in the formula.
- Ignoring leakages: in Cameroon, the cash-drain ratio is historically high (often 30–40%), drastically reducing the real multiplier.

8.3 The Central Bank: BEAC and Monetary Policy in CEMAC

DEFINITION **Central Bank**

A **central bank** is the apex monetary institution entrusted with: (1) issuing legal tender; (2) acting as banker to commercial banks (lender of last resort); (3) acting as banker and fiscal agent to the State; (4) managing foreign exchange reserves; (5) supervising the financial system; (6) conducting monetary policy to achieve price stability and support growth.

PROPERTY **BEAC — Banque des États de l'Afrique Centrale**

- Status: Supranational central bank for the 6 CEMAC member States (Cameroon, Central African Republic, Chad, Congo, Equatorial Guinea, Gabon).
- Headquarters: Yaoundé, Cameroon. National Directorates in each member State.
- Currency: FCFA (Franc de la Coopération Financière en Afrique Centrale), pegged to the euro at 1 EUR = 655.957 FCFA (fixed since 1999).
- Key governance: Monetary Policy Committee (CPM) sets the key rates; COBAC (Commission Bancaire de l'Afrique Centrale) supervises banks and MFIs.
- Convergence criteria (surveillance): Inflation ≤ 3%; Fiscal deficit ≤ 1.5% GDP; Public debt ≤ 70% GDP; Non-accumulation of arrears.

METHOD **Analysing BEAC's Monetary Policy Instruments**

1. Diagnose the macroeconomic situation: inflation above 3% → restrictive policy; recession/unemployment → expansionary policy.
2. Choose the instrument and direction:

Instrument	Restric (Anti-inflatic	Expansionary (Anti-recession)
Key policy rate (TIAO — Taux d'Intérêt des Appels d'Offres)	Raise	Lower
Marginal lending facility rate (TIP — Taux d'Intérêt du Prêt marginal)	Raise	Lower
Reserve requirement ratio (Taux de réserves obligatoires)	Raise (reduces multiplier)	Lower
Open Market Operations (OMO) — BEAC securities	Sell (absorb liquidity)	Buy (inject liquidity)

3. Trace the transmission mechanism:

   Policy rate → Interbank rate → Bank lending rates → Credit volume → Investment/Consumption →

4. Evaluate limits: time lags (6–18 months); imported inflation (oil, food) insensitive to rates; single policy for 6 heterogeneous economies; weak interest-rate pass-through in shallow markets; fiscal dominance.

WORKED EXAMPLE 8 **BEAC Reacts to an Inflationary Shock in CEMAC**

Following a surge in global food and fuel prices, CEMAC inflation rises to 5%, breaching the 3% convergence criterion. The CPM decides to tighten policy.

1. Describe the sequence of instruments BEAC can activate.
2. Trace the transmission to a Cameroonian importer in Douala.
3. Discuss two major limits of this policy in the CEMAC context.

SOLUTION

1. Instruments activated.

- Raise TIAO (e.g. from 3.50% to 4.50%): makes weekly liquidity injections more expensive for banks.

- Raise TIP (marginal lending window): raises the ceiling for overnight borrowing.
- Raise reserve requirement (e.g. from 7% to 9%): directly cuts the credit multiplier $k = 1/r$.
- Open market sales of BEAC bills: mop up excess liquidity permanently.

2. Transmission to a Douala importer. Higher TIAO → banks bid less at auctions → inter-bank rate rises → banks' marginal cost of funds rises → prime lending rate (e.g. from 7.5% to 9%) → the importer's cost of financing letters of credit rises → import volumes contract → demand-pull pressure on prices eases.
3. Limits.
   - Supply-side inflation: the shock comes from global commodity prices and poor domestic logistics (bad roads, port delays). Monetary tightening cannot lower the world price of wheat or fuel.
   - Time lag and fiscal dominance: the policy takes months to bite, while the government may simultaneously run large deficits financed by BEAC advances (statutory advances), injecting liquidity that offsets the tightening.
   - One size fits six: a rate hike suited to Gabon (oil-rich, low inflation) may deepen recession in CAR or Chad.

DID YOU KNOW? **BEAC and the Euro Peg**

The FCFA/euro peg is guaranteed by the French Treasury under a 2019 reform of the co-operation agreements. BEAC must keep at least 50% of its foreign reserves in an operations account at the Banque de France. In return, France guarantees unlimited convertibility. This arrangement delivers exchange-rate stability and low inflation but limits monetary sovereignty: BEAC cannot act as a true lender of last resort in foreign currency, and the fixed parity may become misaligned with fundamentals.

8.4 Microfinance Activities in Cameroon

DEFINITION **Microfinance**

Microfinance refers to the provision of financial services — small loans (**microcredit**), savings, insurance, payment services — to low-income individuals and micro-enterprises excluded from the formal banking sector due to lack of collateral, credit history, or formal employment.

PROPERTY **Microfinance Landscape in Cameroon (COBAC Regulation)**

- Category 1: Networks of credit unions / cooperatives (e.g. **CAMCCUL** — Cameroon Cooperative Credit Union League, **MC²** — Mutuelles Communautaires de Croissance). Member-owned, not-for-profit, mobilise savings from members only.
- Category 2: Microfinance institutions (MFIs) authorised to collect savings from the public and grant credit (e.g. **UCCGN**, **CEC**, **AFG**). For-profit or non-profit, stricter prudential norms.
- Category 3: MFIs authorised to grant credit only (no public savings collection), often funded by donors or wholesale lenders.
- Informal sector: **Tontines** (njangui, esusu) — rotating savings and credit associations (ROSCAs) and accumulating savings and credit associations (ASCAs). Not supervised by COBAC but socially crucial.

METHOD **Analysing the Role and Mechanism of Microfinance**

1. Mechanism: Group lending / solidarity groups → **social collateral** replaces physical collateral; small, frequent repayments (weekly/bi-weekly); compulsory savings build track record; progressive lending (larger loans after successful repayment).
2. Advantages: Financial inclusion (women, youth, rural); poverty reduction; enterprise development; savings mobilisation; women empowerment; social cohesion.
3. Limits/Risks: High effective interest rates (administrative cost of small loans); over-indebtedness (multiple borrowing); weak governance/fraud in some MFIs; mission drift (consumer loans vs productive loans); limited loan sizes constrain growth; vulnerability to covariate shocks (crop failure, pandemic).

WORKED EXAMPLE 9 **Microfinance and a Tomato Trader in Bafoussam**

Mama Ngo, a tomato retailer in Bafoussam market, needs 150 000 FCFA to buy stock. Commercial banks refuse her (no payslip, no land title). Explain how a Category-1 credit union can help her, using the specific mechanisms of microfinance.

SOLUTION

Mechanism at a CAMCCUL-affiliated credit union:

1. Membership & savings: Mama Ngo joins the union, buys a share (e.g. 10 000 FCFA), and starts compulsory weekly savings of 5 000 FCFA. After 3 months, she has 60 000

FCFA savings — this creates a credit history and internal collateral.

2. Loan application: She applies for 150 000 FCFA. The credit committee assesses her character (market reputation), capacity (daily cash flow from tomato sales), capital (her savings), collateral (her savings pledged + two co-signatories from her solidarity group), conditions (working capital, 6-month term).

3. Disbursement & repayment: Loan disbursed directly to her supplier (avoids diversion). Weekly instalments of 6 500 FCFA deducted from daily sales. Group members exert peer pressure for repayment.

Benefits: (i) Access despite no formal collateral; (ii) working capital turnover increases → higher income; (iii) savings habit + financial literacy; (iv) profits reinvested in union → community development.
Risks: (i) Effective rate ≈ 24–30% p.a. (flat rate ≈ 1.5–2%/month) due to high operating costs; (ii) if tomato prices crash (seasonal glut), repayment capacity falls; (iii) risk of borrowing from another MFI simultaneously (multiple indebtedness); (iv) if the credit union fails (governance crisis), her savings are at risk (though COBAC deposit insurance covers up to a limit).
Conclusion: Microfinance fills the missing middle in Cameroon's financial system. Effective supervision (COBAC), credit bureaus (to prevent over-indebtedness), and financial education are essential for sustainability.

8.5 Appraisal of a Loan File by a Credit Union

DEFINITION **Loan Appraisal / Credit Risk Assessment**

Loan appraisal is the systematic evaluation of a borrower's ability and willingness to repay a loan. In microfinance, it combines quantitative (cash flow, ratios) and qualitative (character, social capital) analysis.

METHOD **The 5 C's of Credit Appraisal (Microfinance Adaptation)**

A credit officer must verify each **C** before recommending approval:

1. Character (Intégrité): Honesty, reputation in community, credit history (Credit Bureau / centrale des risques), willingness to repay. Most important C.

2. Capacity (Capacité de remboursement): Ability to generate cash flow to service debt. Compute **Debt Service Ratio (DSR)** = $\frac{\text{Monthly instalment}}{\text{Monthly net cash flow}}$. Threshold: DSR ≤ 30–35% (COBAC guideline). Stress-test with income shocks.

3. Capital (Apport personnel): Borrower's own funds in the project (skin in the game). Reduces moral hazard. Typically ≥ 10–20% of project cost.
4. Collateral (Garanties): Primary: pledged savings, guarantors (caution solidaire), pledge of financed asset. Secondary: moral guarantee, group solidarity. In micro-finance, social collateral often substitutes physical collateral.
5. Conditions (Conditions du prêt): Purpose (productive vs consumption), amount, duration (matching asset life), interest rate, repayment frequency, economic environment (sector outlook, seasonality).

WORKED EXAMPLE 10 **Appraising Mr. Abena's Loan File at a Credit Union**

Mr. Abena, a carpenter in Buea, applies for 1 000 000 FCFA to buy a woodworking machine. File: monthly net income 180 000 FCFA; personal savings at union 200 000 FCFA; two salaried guarantors; two previous loans repaid on time. Proposed loan: 24 months, flat total cost giving monthly instalments of 50 000 FCFA. Appraise using the 5 C's and decide.

SOLUTION

1. Character: Two prior loans repaid fully and on time → excellent credit history. No record at the centrale des risques. Community reputation verified by loan officer visit. Positive.
2. Capacity:

$$\text{DSR} = \frac{50\,000}{180\,000} = 27.8\% < 33\%.$$

Cash flow analysis: daily sales ≈ 15 000 FCFA, material costs ≈ 6 000 FCFA, net daily ≈ 9 000 FCFA → monthly net ≈ 270 000 FCFA (higher than declared). The machine will increase output by ~30%. Positive.
3. Capital: Own savings 200 000 FCFA = 20% of loan amount. Demonstrates commitment. Positive.
4. Collateral: (a) Savings 200 000 FCFA pledged (blocked); (b) Two salaried guarantors (net income > 150 000 FCFA each) signing caution solidaire; (c) Machine to be purchased pledged to union (chattel mortgage). Adequate.
5. Conditions: Purpose: productive asset (machine). Duration: 24 months matches useful life. Rate: flat 2%/month ≈ 24% p.a. effective ≈ 27% — within union's pricing policy. Sector: furniture demand growing in Buea (urbanisation). Positive.
Decision: GRANT the loan with conditions:

- Disbursement directly to machine supplier (pro-forma invoice required).
- Machine insured (fire/theft) with union as beneficiary.
- Quarterly monitoring visits to workshop.
- No further borrowing from other MFIs without union consent (cross-check credit bu-

reau).

Key learning: The 5 C's protect both the borrower (avoids over-indebtedness) and the credit union (preserves members' savings).

WARNING **Examination Traps in This Chapter**

- Balance sheet: A loan granted is an asset (claim on borrower); a deposit received is a liability (obligation to depositor). Never invert them.
- Credit creation: Credit created = Total deposits - Initial deposit. The multiplier $k = 1/r$ applies to the system, not a single bank.
- BEAC: The FCFA/euro peg is fixed by treaty with France, not unilaterally by BEAC. BEAC manages the peg but cannot devalue alone.
- Microfinance categories: Category 1 = member-based credit unions (savings from members only); Category 2 = MFIs taking public savings; Category 3 = credit-only MFIs. Know the difference.
- Loan appraisal: Character comes before Collateral. A borrower with perfect collateral but bad character will default. The 33% DSR is a guideline — always stress-test.
- Terminology: Réserves obligatoires = required reserves (liability side of BEAC, asset side of commercial bank). Réserves (comptables) = retained earnings (equity). Do not confuse.

8.6 Chapter Synthesis: Money and Banking in Cameroon — Key Linkages

KEY POINTS **Essential Linkages for the GCE Exam**

- Commercial banks create credit within the framework set by BEAC (reserve ratio, key rate, liquidity ratio). Their balance sheets reflect this constraint.
- BEAC's monetary policy (TIAO, reserves, OMO) influences the cost and volume of credit → affects investment, consumption, inflation, and the external balance of the CEMAC zone.
- Microfinance complements commercial banks by serving the excluded majority (informal sector, women, rural). It is supervised by COBAC (same supervisor as banks) but operates with different technologies (social collateral, group lending).
- Loan appraisal (5 C's) is the micro-foundation of financial stability: poor appraisal → high NPLs → bank/MFI failure → systemic risk → BEAC/COBAC intervention.

- Structural challenges in Cameroon: High cash-drain ratio, shallow interbank market, dominant informal sector, fiscal dominance, weak collateral registry, low financial literacy — all reduce the effectiveness of monetary policy and the reach of formal finance.

EXERCISE 7 **GCE-Style Practice Questions**

1. Structured question: The balance sheet of Ngounso Bank shows: Cash 2 000; BEAC reserves 5 000; Treasury bills 3 000; Loans 40 000; Fixed assets 10 000; Sight deposits 30 000; Savings deposits 20 000; Time deposits 5 000; Capital 5 000; Reserves 2 000. (All in millions FCFA).
   (a) Present the balance sheet in regulatory format.
   (b) Calculate the liquidity ratio (liquid assets / short-term deposits) and comment on compliance with COBAC norms.
   (c) If BEAC raises the reserve requirement from 7% to 10%, explain the impact on this bank's lending capacity.
2. Essay question: "Monetary policy in the CEMAC zone is ineffective in controlling inflation because inflation is mainly imported and supply-driven." Discuss this statement with reference to BEAC's instruments and the transmission mechanism.
3. Case study: A credit union in Bamenda receives a loan application from a group of 10 women farmers for 5 000 000 FCFA to buy a maize sheller. Outline the appraisal process using the 5 C's, highlighting the specificities of group lending.
4. Calculation: Initial deposit 1 000 million FCFA, reserve ratio 8%, cash-drain ratio 12%, excess reserve ratio 3%. Compute the effective money multiplier, total deposits, and total credit created. Compare with the no-leakage case.

SOLUTION **Exercise 1**

(a) Balance Sheet of Ngounso Bank (millions FCFA)

Assets		Liabilities + Equity	
Cash in vault	2 000	Share capital	5 000
Reserves at BEAC	5 000	Reserves (retained profits)	2 000
Treasury bills	3 000	Time deposits	5 000
Loans to customers	40 000	Savings deposits	20 000
Fixed assets	10 000	Sight deposits	30 000
Total	60 000	Total	60 000

(b) Liquidity ratio: Liquid assets = 2 000 + 5 000 + 3 000 = 10 000. Short-term deposits = 30 000 + 20 000 + 5 000 = 55 000. Ratio = 10 000 / 55 000 = 18.2%. Non-compliant with COBAC 100% norm. The bank is highly illiquid and depends on BEAC refinancing.
(c) Impact of reserve ratio increase: Required reserves rise from 7% × 55 000 = 3 850 to 10% × 55 000 = 5 500. The bank currently holds 5 000 at BEAC. It must either borrow 500 on the interbank market, sell Treasury bills, or reduce new lending. Its lending capacity shrinks.

SOLUTION **Exercise 4**

No leakages: $k = 1/0.08 = 12.5$; $D = 12\,500$; $C = 11\,500$. With leakages: $k_{eff} = 1/(0.08+0.12+0.03) = 1/0.23 \approx 4.348$; $D' \approx 4\,348$; $C' \approx 3\,348$. Leakages reduce the multiplier by a factor of ≈ 2.87, illustrating the huge impact of the informal cash economy and banks' caution in Cameroon.

9 EXERCISES

9.1 I check my knowledge

EXERCISE 8 **MCQ on Money and Banking Fundamentals**

Answer the following multiple choice questions by selecting the correct option.

1. Which of the following is a liability on a commercial bank's balance sheet?

   A. Cash in vault

   B. Loans granted to customers

   C. Customer deposits

   D. Treasury bills held

2. The credit multiplier (K) is correctly given by which formula, where r is the reserve ratio?

   A. $K = r$

   B. $K = \frac{1}{r}$

   C. $K = 1 - r$

   D. $K = \frac{r}{1-r}$

3. The Bank of Central African States (BEAC) acts as the central bank for which monetary zone?

   A. WAEMU (UEMOA)

   B. CEMAC

   C. ECOWAS (CEDEAO)

   D. SADC

4. Which institution is responsible for the supervision and classification of Microfinance Institutions (MFIs) in the CEMAC zone?

   A. BEAC

   B. COBAC

   C. IMF

   D. Ministry of Finance

5. In the 5 Cs of credit appraisal, which "C" refers to the borrower's reputation and track record of repaying debts?

A. Capacity
B. Capital
C. Character
D. Collateral

EXERCISE 9 **True/False with Justification**

State whether each statement is True or False. Justify your answer in one or two sentences.

1. Commercial banks create money by lending out their excess reserves.
2. A higher reserve ratio (r) leads to a higher credit multiplier and more money creation.
3. The TIAO (Taux d'Intérêt des Appels d'Offres) is the rate at which BEAC lends to commercial banks through open market operations.
4. Microfinance Institutions in Cameroon are allowed to collect savings from the public only if they are classified in Category 1 by COBAC.
5. "Collateral" in the 5 Cs framework refers exclusively to land titles owned by the borrower.

EXERCISE 10 **Key Definitions**

Define the following terms precisely, as required in the GCE Advanced Level Economics examination.

1. Balance sheet of a commercial bank
2. Credit creation (money multiplier process)
3. Legal reserve requirement (reserve ratio)
4. Central Bank (with reference to BEAC)
5. Microfinance Institution (MFI) – Category 2 (COBAC classification)
6. The 5 Cs of credit appraisal

EXERCISE 11 **Direct Calculations: Credit Multiplier and Money Creation**

Assume the banking system in Cameroon has an initial deposit of 2 000 000 FCFA and a legal reserve ratio of 10%.

1. Calculate the credit multiplier (K).
2. Calculate the maximum total deposits that can be created by the banking system.
3. Calculate the maximum total new credit (loans) that can be generated.
4. Recalculate (i), (ii) and (iii) if the reserve ratio is increased to 20%.
5. Briefly comment on the effect of changing the reserve ratio on the money supply.

9.2 I practise

EXERCISE 12 **Constructing a Commercial Bank Balance Sheet**

The following items (in millions of FCFA) are extracted from the books of Société Générale des Banques au Cameroun (SGBC) at the end of the financial year:

Cash in vault	1 200
Reserves at BEAC	3 500
Treasury Bills (short-term)	2 800
Loans granted to customers	15 000
Overdrafts	4 200
Buildings and equipment	5 000
Customer deposits: Current accounts	12 000
Customer deposits: Savings accounts	8 000
Customer deposits: Term deposits	5 000
Borrowed funds (from BEAC/other banks)	3 000
Equity (Share capital + Reserves)	3 700

1. Construct the complete Balance Sheet of SGBC in the standard format (Assets on the left, Liabilities & Capital on the right). Ensure it balances.
2. Calculate the Liquidity Ratio defined as:

$$\text{Liquidity Ratio} = \frac{\text{Liquid Assets (Cash + Reserves at BEAC + Treasury Bills)}}{\text{Total Customer Deposits}} \times 100\%$$

3. Comment on the liquidity position of SGBC if the regulatory minimum liquidity ratio set by COBAC is 30%.

EXERCISE 13 **Credit Creation Process: Numerical Application**

Consider a simplified banking system with only one commercial bank (or a system where all banks operate identically). The legal reserve ratio (r) is 10%. A customer deposits 5 000 000 FCFA in cash at the bank.

1. Show the changes in the bank's balance sheet after the initial deposit (before any lending).
2. The bank lends out its maximum excess reserves. Show the balance sheet after the first loan is granted and the borrower writes a cheque that is deposited in the same bank (or another bank in the system).
3. Continue the process for two more rounds of lending and re-depositing. Present a summary table showing: Round, New Deposit, Required Reserves, Excess Reserves (New Loans), Cumulative Deposits, Cumulative Loans.
4. Using the credit multiplier formula, state the theoretical maximum total deposits, total loans, and total money supply created from the initial 5 000 000 FCFA.
5. Explain why the actual money creation in Cameroon might be lower than this theoretical maximum. Give three reasons.

EXERCISE 14 **Microfinance Institutions vs Commercial Banks**

Answer the following short-answer questions.

1. Distinguish between a Commercial Bank and a Microfinance Institution (MFI) in Cameroon using four criteria: (a) Legal framework/Supervisory body, (b) Target clientele, (c) Range of services, (d) Capital requirements.
2. State three specific financial services typically offered by MFIs in Cameroon (e.g., UCC, MC2) that are tailored to the informal sector or low-income households.
3. Explain why the BEAC raising the TIAO (Taux d'Intérêt des Appels d'Offres) tends to reduce inflation in the CEMAC zone. Use the transmission mechanism: TIAO → Bank lending rates → Investment/Consumption → Aggregate Demand → Price Level.

EXERCISE 15 **Loan Appraisal using the 5 Cs – Practical Case**

A farmer, Mr. Nfor, applies for a loan of 600 000 FCFA from a Credit Union (Category 2 MFI) to purchase a cocoa dryer and improve his fermentation process. He submits the following information:

- Monthly net income from cocoa sales: 120 000 FCFA (seasonal, high Oct–Dec, low

Jan–Sep).

- Monthly household expenses: 70 000 FCFA.
- He owns the cocoa farm (5 hectares) but has no land title (customary land tenure).
- He provides two guarantors: a civil servant (salary 250 000 FCFA/month) and a trader (estimated income 180 000 FCFA/month).
- He has been a member of the Credit Union for 3 years, with savings of 150 000 FCFA and a previous loan of 200 000 FCFA repaid on schedule.
- The dryer costs 550 000 FCFA; installation and training 50 000 FCFA.

Appraise this loan file using the 5 Cs framework (Character, Capacity, Capital, Collateral, Conditions). For each "C", state your assessment (Strong / Adequate / Weak) and justify with data from the case. Conclude with a recommendation: Approve, Approve with conditions, or Reject.

9.3 I prepare for the GCE

EXERCISE 16 **Essay: Functions of BEAC as Central Bank – 25 marks**

"Explain the functions of a central bank, illustrating each with the activities of the Bank of Central African States (BEAC) in the CEMAC zone."
Guidance for structuring your answer (not part of the exam question):

- Introduction: Define central bank; state BEAC's role in CEMAC (6 member states, common currency CFA Franc).
- Function 1: Monetary Policy / Price Stability (TIAO, reserve requirements, open market operations).
- Function 2: Banker to the Government (managing public debt, Treasury accounts – note CEMAC rules on advances).
- Function 3: Banker to Commercial Banks (lender of last resort, clearing house, holding reserves).
- Function 4: Issuer of Currency (monopoly on banknotes/coins, managing foreign exchange reserves).
- Function 5: Supervision & Regulation (with COBAC – prudential ratios, licensing).
- Function 6: Manager of Foreign Exchange Reserves (operations account with French Treasury, exchange rate peg).
- Conclusion: Summary of BEAC's effectiveness / challenges (e.g., asymmetric

shocks, limited policy autonomy).

EXERCISE 17 **Essay: Problems of Commercial Banks in Cameroon – 25 marks**

"Discuss the problems faced by commercial banks in Cameroon and suggest possible solutions."
Guidance for structuring your answer (not part of the exam question):

- Introduction: Importance of commercial banks in financial intermediation; context of Cameroonian banking sector (oligopoly, foreign-owned dominance).
- Problem 1: High Non-Performing Loans (NPLs) – causes (poor appraisal, weak legal enforcement, economic shocks).
- Problem 2: Excess Liquidity / Lack of viable projects – paradox of high liquidity coexisting with credit rationing for SMEs.
- Problem 3: Weak Judicial System / Collateral enforcement difficulties (land title issues, lengthy court procedures).
- Problem 4: High Operating Costs / Infrastructure deficits (electricity, transport, ICT) reducing profitability.
- Problem 5: Regulatory Burden / Compliance costs (COBAC Basel II/III implementation, AML/CFT).
- Problem 6: Competition from Mobile Money / Fintech / MFIs eroding traditional deposit base.
- Solutions: For each problem, propose a concrete policy or bank-level solution (e.g., credit bureaus, movable collateral registry, specialized SME desks, digital transformation, public-private partnerships).
- Conclusion: Balanced assessment – progress made (e.g., CEMAC banking reforms) vs persistent structural issues.

EXERCISE 18 **Case Study: Credit Union Loan Appraisal – 20 marks**

Case Study: Mrs. Aisha runs a small food processing cooperative (cassava flour, "garri") in a rural area of the East Region. She applies for a loan of 1 500 000 FCFA from a Category 2 MFI (Credit Union) to buy a mechanical grater, a hydraulic press, and build a drying shed. The total project cost is 1 800 000 FCFA; she contributes 300 000 FCFA from own savings.
File Summary:

- Character: Member for 5 years; savings 400 000 FCFA; two previous loans (500 000 FCFA and 800 000 FCFA) repaid early. Treasurer of her church women's group. No credit bureau record (rural area).
- Capacity: Cooperative monthly revenue 1 200 000 FCFA; operating costs 700 000 FCFA (raw material, labour, transport); net surplus 500 000 FCFA/month. Loan repayment proposed: 100 000 FCFA/month over 18 months. Household expenses 150 000 FCFA/month.
- Capital: Own contribution 300 000 FCFA (16.7% of project cost). Cooperative assets (equipment, stock) valued at 2 000 000 FCFA.
- Collateral: No land title for the shed location (community land). Proposes: (a) Chattel mortgage on the new equipment (grater, press) valued at 1 200 000 FCFA; (b) Personal guarantee from the Cooperative President (civil servant, salary 300 000 FCFA); (c) Pledge of her savings 400 000 FCFA in the Credit Union.
- Conditions: Growing local demand for packaged garri; new rural market opening 5 km away; input prices (cassava) stable; no major competitors. Risk: Seasonal cassava shortage (Jan–Mar), poor road access during rains.

Tasks:

1. Appraise the loan file using the 5 Cs. For each C, provide a rating (Strong / Adequate / Weak) and a detailed justification referencing the data above. [10 marks]
2. Calculate the Debt Service Coverage Ratio (DSCR) for the cooperative: $\frac{\text{Net Surplus}}{\text{Proposed Monthly Repayment}}$. Interpret the result. [3 marks]
3. As the Credit Committee, state your final decision: Approve / Approve with Conditions / Reject. If "Approve with Conditions", specify three binding conditions (covenants) to mitigate the identified risks. [7 marks]

10 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ on Money and Banking Fundamentals**

1. Correct answer: C — Customer deposits.

   Justification: On a commercial bank's balance sheet, the liabilities side records everything the bank owes. Customer deposits are debts of the bank towards its clients, who can withdraw them on demand (current accounts) or after notice (savings and term accounts). Options A, B and D are assets (uses of funds): cash in vault is a liquid asset, loans granted are claims on customers, and Treasury bills are financial investments.

2. Correct answer: B — $K = \frac{1}{r}$.

   Justification: In the simplified credit-creation model (no cash leakage, banks lend all excess reserves), the credit multiplier is the inverse of the legal reserve ratio. With $r = 10\%$, $K = 10$: each franc of new deposit can support ten francs of total deposits in the system. Option A confuses the multiplier with the ratio itself; C and D have no economic meaning here.

3. Correct answer: B — CEMAC.

   Justification: The BEAC (Bank of Central African States) is the common central bank of the six member states of the Communauté Économique et Monétaire de l'Afrique Centrale: Cameroon, Chad, Central African Republic, Congo, Equatorial Guinea and Gabon, which share the CFA Franc (XAF). WAEMU is served by the BCEAO; ECOWAS and SADC are not monetary unions with a single central bank of this type.

4. Correct answer: B — COBAC.

   Justification: The Commission Bancaire de l'Afrique Centrale (COBAC) is the CEMAC banking supervisory authority. It licenses, classifies (Categories 1, 2 and 3) and supervises microfinance institutions, in addition to supervising commercial banks. The BEAC conducts monetary policy; the Ministry of Finance handles State finances; the IMF is an external institution.

5. Correct answer: C — Character.

   Justification: Character assesses the borrower's honesty, reputation and past repayment behaviour (credit history, membership record in the Credit Union). Capacity is the ability to repay from income; Capital is the borrower's own financial contribution; Collateral is the security pledged against the loan; Conditions refer to the purpose of the loan and the economic environment.

WARNING **Typical MCQ traps**

Candidates frequently confuse assets and liabilities (deposits "belong" to customers, hence they are a liability for the bank), and confuse BEAC (monetary policy) with COBAC (banking supervision). Always ask: "who owes what to whom?" and "who does what?".

SOLUTION **Exercise 2 — True/False with Justification**

1. TRUE. When a bank's actual reserves exceed its legal (required) reserves, it can grant new loans by simply crediting the borrower's account; this new deposit is new (scriptural) money. Money is destroyed symmetrically when loans are repaid.
2. FALSE. The credit multiplier is $K = \frac{1}{r}$, so it varies inversely with the reserve ratio. A higher r forces banks to immobilise more reserves per franc of deposit, leaving less to lend: the multiplier falls (e.g. $r = 10\% \Rightarrow K = 10$; $r = 20\% \Rightarrow K = 5$).
3. TRUE. The TIAO (Taux d'Intérêt des Appels d'Offres) is BEAC's key policy rate: the rate at which commercial banks obtain central bank refinancing through weekly tenders (open market operations). Changes in the TIAO are transmitted to banks' lending rates and hence to economic activity and prices.
4. TRUE. Under the COBAC classification, only Category 1 MFIs (savings and credit cooperatives such as Credit Unions and MC^2) are authorised to collect savings/deposits from their members and from the public within regulatory limits. Category 2 institutions grant credit only (no deposit collection), and Category 3 covers other limited operators.
5. FALSE. Collateral covers any asset pledged to secure a loan: land titles and buildings (mortgages), but also vehicles, equipment (chattel mortgage), stocks of goods, savings pledged in the institution, and even personal guarantees (guarantors). Restricting collateral to land titles is precisely one of the obstacles to credit access in Cameroon, where many borrowers hold land under customary tenure without formal titles.

SOLUTION **Exercise 3 — Key Definitions**

1. Balance sheet of a commercial bank: A financial statement, at a given date, showing the bank's assets (uses of funds: cash, reserves at the central bank, securities, loans

and overdrafts, fixed assets) on the left and its liabilities and equity (sources of funds: customer deposits, borrowed funds, share capital and reserves) on the right, with the two sides always equal: Total Assets = Total Liabilities + Equity.

2. Credit creation (money multiplier process): The process by which the banking system as a whole creates scriptural money by granting loans: an initial deposit generates excess reserves, which are lent and re-deposited repeatedly, so that total deposits equal the initial deposit multiplied by the credit multiplier $K = \frac{1}{r}$, where r is the reserve ratio.

3. Legal reserve requirement (reserve ratio): The minimum fraction r of customer deposits that every commercial bank is obliged by the monetary authority (BEAC/COBAC) to hold as reserves (in vault cash or on its account at the central bank) rather than lend out. It is an instrument of monetary policy and of banking security.

4. Central Bank (with reference to BEAC): The public monetary institution that holds the monopoly of currency issue, conducts monetary policy, acts as banker to the State and to commercial banks (lender of last resort), manages foreign exchange reserves and (with COBAC) supervises the banking system. The BEAC fulfils these functions for the six CEMAC states, issuing the CFA Franc (XAF) from its headquarters in Yaoundé.

5. Microfinance Institution — Category 2 (COBAC): A financial institution, licensed and supervised by COBAC, authorised to grant microcredit to low-income clients and the informal sector but not authorised to collect savings/deposits from the public; it finances its lending from own funds and borrowing. (Category 1 = savings and credit cooperatives collecting deposits; Category 3 = other limited operators.)

6. The 5 Cs of credit appraisal: The five criteria used by a lender to evaluate a loan file: Character (honesty, repayment track record), Capacity (ability to repay from income/cash flow), Capital (borrower's own contribution and net worth), Collateral (security pledged) and Conditions (purpose of the loan and the economic environment of the activity).

METHOD **Writing a good GCE definition**

A full-mark definition contains: (i) the genus (what kind of thing it is — a statement, a process, a ratio, an institution); (ii) the essential characteristics (what distinguishes it); (iii) where useful, a formula or an example. One or two precise sentences suffice — do not write a paragraph of vague description.

SOLUTION **Exercise 4 — Direct Calculations: Credit Multiplier and Money Creation**

Given: initial deposit D_0 = 2 000 000 FCFA; reserve ratio r = 10% = 0.10.

1. Credit multiplier:
$$K = \frac{1}{r} = \frac{1}{0.10} = 10$$
2. Maximum total deposits:
$$D_{max} = K \times D_0 = 10 \times 2\,000\,000 = 20\,000\,000 \text{ FCFA}$$
3. Maximum total new credit (loans): Total loans equal total deposits minus the initial deposit (which is "old" money, not created):
$$C_{max} = D_{max} - D_0 = 20\,000\,000 - 2\,000\,000 = 18\,000\,000 \text{ FCFA}$$
(Equivalently, $C_{max} = D_0 \times \frac{1-r}{r} = 2\,000\,000 \times 9$.)
4. With r = 20% = 0.20:
$$K = \frac{1}{0.20} = 5$$
$$D_{max} = 5 \times 2\,000\,000 = 10\,000\,000 \text{ FCFA}$$
$$C_{max} = 10\,000\,000 - 2\,000\,000 = 8\,000\,000 \text{ FCFA}$$
5. Comment: Doubling the reserve ratio from 10% to 20% halves the multiplier (from 10 to 5) and therefore halves the money-creation capacity of the banking system (deposits fall from 20 to 10 million FCFA; new credit from 18 to 8 million FCFA). The reserve ratio is thus a powerful contractionary/expansionary instrument: raising r reduces the money supply (fighting inflation), while lowering r expands credit (stimulating activity).

WARNING **Common mistake**

Do not confuse total deposits ($K \times D_0$) with total new credit ($K \times D_0 - D_0$). The initial deposit is not "created" money; examiners deduct marks when candidates present 20 million as "credit created".

SOLUTION **Exercise 5 — Constructing a Commercial Bank Balance Sheet**

1. Balance sheet of SGBC (millions of FCFA). Classification: cash, reserves at BEAC, Treasury bills, loans, overdrafts and buildings are assets; deposits and borrowed funds are liabilities; equity is capital.

ASSETS	Amount	LIABILITIES & CAPITAL	Amount
Cash in vault	1 200	Current account deposits	12 000
Reserves at BEAC	3 500	Savings account deposits	8 000
Treasury Bills	2 800	Term deposits	5 000
Loans to customers	15 000	Borrowed funds (BEAC/banks)	3 000
Overdrafts	4 200	Equity (capital + reserves)	3 700
Buildings and equipment	5 000		
TOTAL ASSETS	31 700	TOTAL LIAB. + CAPITAL	31 700

Check: Assets: 1 200+3 500+2 800+15 000+4 200+5 000 = 31 700. Liabilities + capital: 12 000 + 8 000 + 5 000 + 3 000 + 3 700 = 31 700. The balance sheet balances. ■

2. Liquidity Ratio:

$$\text{Liquid assets} = 1\,200 + 3\,500 + 2\,800 = 7\,500 \text{ (millions FCFA)}$$

$$\text{Total customer deposits} = 12\,000 + 8\,000 + 5\,000 = 25\,000 \text{ (millions FCFA)}$$

$$\text{Liquidity Ratio} = \frac{7\,500}{25\,000} \times 100\% = 30\%$$

3. Comment: SGBC's liquidity ratio (30%) is exactly equal to the COBAC regulatory minimum of 30%. The bank is therefore just compliant: it holds enough liquid assets to meet the prudential requirement, but with no safety margin. Any unexpected deposit withdrawal, or any fall in the value of its Treasury bills, would push the bank below the minimum and expose it to COBAC sanctions or forced refinancing at BEAC (at the TIAO or penalty rate). Prudent management would target a ratio above the minimum (e.g. 33–35%), even though holding more low-yield liquid assets reduces profitability — the classic liquidity–profitability trade-off.

SOLUTION **Exercise 6 — Credit Creation Process: Numerical Application**

Given: r = 10%; initial cash deposit D_0 = 5 000 000 FCFA.

1. Balance sheet after the initial deposit (before lending):

ASSETS	FCFA	LIABILITIES	FCFA
Cash / Reserves	5 000 000	Demand deposit (Customer A)	5 000 000
Total	5 000 000	Total	5 000 000

Required reserves: 0.10 × 5 000 000 = 500 000; excess reserves = 5 000 000 − 500 000 = 4 500 000.

2. After the first loan (4 500 000 lent, then re-deposited by the cheque recipient):

ASSETS	FCFA	LIABILITIES	FCFA
Reserves	5 000 000	Deposit A	5 000 000
Loan to B	4 500 000	Deposit (recipient of B's cheque)	4 500 000
Total	9 500 000	Total	9 500 000

New required reserves: 0.10 × 9 500 000 = 950 000; excess reserves = 5 000 000 − 950 000 = 4 050 000, which can be lent in round 3.

3. Summary table (rounds 1 to 4):

Round	New deposit	Required reserves (10%)	New loan (excess)	Cumul. deposits	Cumul. loans
1	5 000 000	500 000	4 500 000	5 000 000	0
2	4 500 000	450 000	4 050 000	9 500 000	4 500 000
3	4 050 000	405 000	3 645 000	13 550 000	8 550 000
4	3 645 000	364 500	3 280 500	17 195 000	12 195 000

Each new deposit is 90% of the previous one: the process is a geometric progression of ratio $(1 - r) = 0.9$.

4. Theoretical maximum (multiplier formula):

$$K = \frac{1}{r} = \frac{1}{0.10} = 10$$

D_{max} = 10 × 5 000 000 = 50 000 000 FCFA (total deposits)

C_{max} = 50 000 000 − 5 000 000 = 45 000 000 FCFA (total loans)

Money created = 45 000 000 FCFA of new scriptural money.

5. Why actual money creation in Cameroon is lower — three reasons:

- Cash leakages: in a largely cash-based, informal economy, borrowers withdraw a significant part of loans in banknotes instead of re-depositing them; each leakage withdraws reserves from the banking circuit and shrinks the multiplier.
- Banks do not lend all excess reserves: faced with high non-performing loans and few "bankable" projects, Cameroonian banks voluntarily hold excess liquidity (or buy Treasury bills) rather than lend, especially to SMEs.
- Limited demand for credit / bankable demand: many potential borrowers lack land titles, guarantors or formal accounts, so they cannot obtain loans; without new loans, the multiplication process stops. (One may also cite: prudential liquidity ratios imposed by COBAC above the legal reserve, and imports/payments abroad that drain reserves out of the domestic system.)

SOLUTION **Exercise 7 — Microfinance Institutions vs Commercial Banks**

1. Four distinguishing criteria:

Criterion	Commercial Bank	Microfinance Institution
(a) Legal framework / supervision	Banking law of CEMAC; licensed and supervised by COBAC as credit establishments; large minimum capital (billions of FCFA).	Specific CEMAC microfinance regulation (COBAC classification, Categories 1–3); often cooperative law (OHADA / COBAC); much lower minimum capital.
(b) Target clientele	Formal sector: companies, civil servants, salaried workers, traders with guarantees and land titles.	Informal sector and low-income households: farmers, petty traders, artisans, women's groups, rural populations.
(c) Range of services	Full range: current accounts, cheques, overdrafts, international transfers, trade finance, cards, foreign exchange.	Narrow range: member savings, small loans (microcredit), group lending, sometimes money transfer; no cheques or foreign operations.
(d) Capital requirements	High minimum share capital fixed by COBAC/CEMAC banking regulation.	Low minimum capital adapted to cooperative size; funds come mainly from members' shares and savings.

2. Three services tailored to the informal sector (e.g. UCC, MC^2):

- Microcredit / small seasonal loans without land-title collateral (e.g. loans to farmers for inputs, repaid at harvest), often with guarantors or group guarantee instead.
- Mobilisation of small, flexible savings (daily or weekly deposits, "susu"-type collection), accessible in neighbourhoods and villages where banks have no branches.
- Group (solidarity) lending and financial education: loans to solidarity groups (e.g. njangui/tontine members) where members guarantee each other, plus training in basic management of income and repayment discipline.

3. Transmission mechanism of a TIAO increase:

- TIAO ↑: refinancing at BEAC becomes more expensive for commercial banks (their cost of funds rises).
- → Bank lending rates ↑: banks pass the higher cost onto customers; credit also becomes scarcer as banks' liquidity tightens.
- → Investment and consumption ↓: firms borrow less for investment; households reduce credit-financed consumption.
- → Aggregate demand ↓: total spending in the CEMAC economy contracts relative to supply.
- → Price level: demand-pull inflationary pressure eases, so the general price level stabilises (inflation falls).

This is the standard restrictive monetary policy channel through which BEAC defends price stability and the internal/external value of the CFA Franc.

SOLUTION **Exercise 8 — Loan Appraisal using the 5 Cs (Mr. Nfor)**

Loan requested: 600 000 FCFA (dryer 550 000 + installation/training 50 000 — the budget is coherent and fully costed).

1. Character — STRONG. Member for 3 years; savings of 150 000 FCFA; a previous loan of 200 000 FCFA repaid on schedule. This demonstrates reliability and attachment to the Credit Union.
2. Capacity — ADEQUATE (with reservation). Net monthly income 120 000 FCFA minus household expenses 70 000 FCFA leaves a surplus of about 50 000 FCFA/month, i.e. 600 000 FCFA/year — sufficient to repay 600 000 FCFA over roughly 12–18 months with interest. However, income is highly seasonal (high Oct–Dec, low Jan–Sep): repayment must be scheduled on the cocoa season, not in equal monthly instalments. The dryer should also raise income (better-fermented, well-dried cocoa sells at a higher price), improving future capacity.
3. Capital — ADEQUATE. Own savings of 150 000 FCFA represent 25% of the loan amount, and he owns a productive 5-hectare farm. He has a real stake in the project, though his cash contribution to the project itself is modest.
4. Collateral — WEAK to ADEQUATE. No land title (customary tenure) means no mortgage is possible — a classic Cameroonian constraint. Mitigating elements: two guarantors, including a civil servant with a stable salary of 250 000 FCFA/month (a strong guarantee) and a trader (180 000 FCFA/month); plus the possibility of pledging his 150 000 FCFA savings and taking a chattel mortgage on the dryer. The guarantee package partly compensates the absence of a land title.
5. Conditions — STRONG. The purpose is productive (equipment that improves quality

and value added, not consumption); cocoa is a strategic Cameroonian export crop with steady demand; the total cost (600 000 FCFA) matches the quotation exactly.

Recommendation: APPROVE WITH CONDITIONS.

- Repayment schedule indexed on the cocoa season (larger instalments Oct–Dec, token payments off-season).
- Pledge of the 150 000 FCFA savings + signed guarantees of the two guarantors (civil servant first) + chattel mortgage on the dryer.
- Disbursement directly to the equipment supplier (not in cash to the borrower) to guarantee the use of funds.

SOLUTION **Exercise 9 — Essay: Functions of BEAC as Central Bank (25 marks)**

Indicative mark scheme (examiner expectations):

Component	Marks
Introduction: definition of a central bank; BEAC, CEMAC (6 states), CFA Franc	3
Function 1: Monetary policy / price stability (TIAO, reserve requirements, open market)	4
Function 2: Banker to the Government (Treasury accounts, public debt; statutory limits on advances)	3
Function 3: Banker to commercial banks (refinancing, lender of last resort, clearing, reserves)	4
Function 4: Issuer of currency (monopoly of issue of notes and coins)	3
Function 5: Supervision and regulation (with COBAC: licensing, prudential ratios)	3
Function 6: Management of foreign exchange reserves (peg to the euro, operations account with the French Treasury)	3
Conclusion: balanced assessment (effectiveness, challenges)	2
Total	25

Model answer (outline developed):
Introduction. A central bank is the public institution at the apex of a country's monetary and banking system, responsible for issuing currency, conducting monetary policy and safeguarding financial stability. For the six CEMAC states (Cameroon, Chad, Central African Republic, Congo, Equatorial Guinea, Gabon), this role is performed collectively by the Bank of Central African States (BEAC), headquartered in Yaoundé, which issues the common currency, the CFA Franc (XAF).
1. Conduct of monetary policy and price stability. BEAC's primary mission is price stability in the zone. It fixes its key policy rate, the TIAO (rate of its weekly liquidity tenders),

sets legal reserve requirements for commercial banks, and conducts open market operations. Raising the TIAO makes bank refinancing dearer, restrains credit and cools inflation; lowering it stimulates activity.
2. Banker to the Government. BEAC keeps the accounts of the member States' Treasuries, manages their public debt issuance (Treasury bills and bonds on the CEMAC market) and may grant advances within strict statutory limits — a discipline designed to prevent the inflationary monetary financing of budget deficits that marked the region in the past.
3. Banker to commercial banks. Commercial banks hold their required reserves at BEAC, obtain refinancing there (BEAC is lender of last resort to solvent but illiquid banks), and settle interbank payments through BEAC's clearing and payment systems. BEAC is thus the hub of the zone's banking circuit.
4. Issuer of currency. BEAC holds the monopoly of issuing banknotes and coins in CEMAC. It guarantees the quality and confidence in the currency, replaces worn notes and combats counterfeiting.
5. Supervision and regulation. Together with COBAC (the banking commission), BEAC participates in licensing banks and MFIs and enforcing prudential rules (capital adequacy, liquidity ratios, classification of microfinance institutions), protecting depositors and systemic stability.
6. Manager of foreign exchange reserves. BEAC centralises and manages the zone's external reserves, which back the fixed peg of the CFA Franc to the euro, guaranteed by the French Treasury through the operations account mechanism, in exchange for the pooling of reserves and convertibility rules.
Conclusion. BEAC performs all the classical functions of a central bank, but in a multinational framework: this guarantees monetary credibility and a stable, convertible currency, at the cost of limited national policy autonomy and difficulties in responding to asymmetric shocks (e.g. an oil-price fall hitting Congo differently from Cameroon). Recent CEMAC reforms aim to strengthen its independence, transparency and supervisory effectiveness.
Examiner expectations: each function must be (i) named, (ii) explained generally, and (iii) illustrated with a BEAC-specific instrument or fact (TIAO, COBAC, operations account, Yaoundé headquarters). A purely theoretical answer with no BEAC illustration loses the application marks.

SOLUTION **Exercise 10 — Essay: Problems of Commercial Banks in Cameroon (25 marks)**

Indicative mark scheme:

Component	Marks
Introduction: role of banks in intermediation; structure of the Cameroonian sector	3
Problem 1: High non-performing loans (causes, consequences)	3
Problem 2: Excess liquidity paradox / credit rationing of SMEs	3
Problem 3: Weak collateral enforcement and judicial delays	3
Problem 4: High operating costs / infrastructure deficits	2
Problem 5: Regulatory and compliance burden	2
Problem 6: Competition from mobile money, fintech and MFIs	2
Solutions matched to problems (at least 4 concrete solutions)	5
Conclusion: balanced assessment	2
Total	25

Model answer (developed outline):
Introduction. Commercial banks are the engine of financial intermediation: they collect savings and transform them into credit for investment and growth. In Cameroon the sector is an oligopoly dominated by a few large, mostly foreign-owned banks (Afriland, SGBC, BICEC, Ecobank, SGC…), alongside MFIs and mobile money operators. Despite substantial resources, these banks face serious problems that limit their contribution to development.
Problem 1 — High non-performing loans (NPLs). A significant share of loans is not repaid, because of poor ex-ante appraisal, weak borrower information, economic shocks (fall in cocoa/oil prices, crises in the North-West/South-West and Far North) and deliberate default. NPLs erode bank profits and capital, and make banks excessively cautious.
Problem 2 — The excess liquidity paradox. Cameroonian banks are highly liquid, yet SMEs and farmers complain of credit rationing. Banks prefer risk-free Treasury bills and lending to the State or large corporations, because small borrowers lack guarantees, formal accounts and viable projects. Liquidity coexists with under-financing of the real economy.
Problem 3 — Collateral and judicial weaknesses. Most rural land is held under customary tenure without land titles, so it cannot be mortgaged; and when default occurs, court procedures to seize and sell collateral are slow and uncertain. Risk therefore appears high, and interest rates and guarantee requirements rise accordingly.
Problem 4 — High operating costs. Electricity cuts, poor roads, costly ICT and security, and the need for own generators raise operating costs, compress margins and keep bank charges and lending rates high.
Problem 5 — Regulatory and compliance burden. Implementing COBAC's prudential framework (Basel-type capital rules), anti-money-laundering and counter-terrorism-financing obligations impose heavy compliance costs, especially on smaller institutions.
Problem 6 — New competition. Mobile money (MTN MoMo, Orange Money), fintechs and MFIs capture deposits and payment services, particularly among the young and the informal sector, eroding the traditional deposit base of banks.

Solutions.

- Against NPLs and information asymmetry: operational credit bureaus and a shared database of borrowers; better staff training in the 5 Cs appraisal.
- Against the collateral problem: a registry of movable collateral (equipment, stocks, receivables) under OHADA law; faster commercial courts and arbitration; land-tenure reform to expand titling.
- To finance SMEs: specialised SME desks, partial public guarantee funds, and channelling excess liquidity into productive credit through refinancing windows.
- To cut costs and face fintechs: digital transformation (agency banking, mobile apps, partnerships with mobile money operators rather than confrontation).
- Public action: investment in electricity and transport infrastructure; proportionate, stable regulation to avoid unnecessary compliance costs.

Conclusion. CEMAC banking reforms (stronger COBAC supervision, higher capital requirements, modernised payment systems) have stabilised the sector, but structural problems — NPLs, collateral enforcement, SME exclusion — persist. Solving them requires joint action by banks (better appraisal, innovation), the State (justice, land reform, infrastructure) and the regulator (proportionate, incentive-based rules), so that the sector's abundant liquidity finally irrigates the Cameroonian economy.
Examiner expectations: a problem must always be explained (cause → mechanism → consequence), not merely listed; solutions must be matched to the problems; Cameroonian/CEMAC illustrations earn the application marks.

SOLUTION **Exercise 11 — Case Study: Credit Union Loan Appraisal (20 marks)**

1. Appraisal with the 5 Cs [10 marks — 2 per C: rating + justification].

- Character — STRONG. Five years of membership; 400 000 FCFA of savings; two previous loans (500 000 FCFA and 800 000 FCFA) repaid early; treasurer of her church women's group (community trust). The absence of a credit-bureau record is neutral in a rural area — her internal track record is excellent.
- Capacity — STRONG. Net cooperative surplus: 1 200 000 - 700 000 = 500 000 FCFA/month. Even after household expenses (150 000 FCFA), about 350 000 FCFA/month remains available, against a proposed repayment of 100 000 FCFA/month. Repayment absorbs only 20% of the net surplus — very comfortable.
- Capital — ADEQUATE. Own contribution of 300 000 FCFA, i.e. $\frac{300\,000}{1\,800\,000}$ = 16.7% of project cost — acceptable (many MFIs require 10–20%), though not high. Cooperative assets of 2 000 000 FCFA exceed the loan, showing a solid patrimony.

- Collateral — ADEQUATE. No land title (community land) — the classic weakness. But the package is reasonable: chattel mortgage on equipment worth 1 200 000 FCFA (80% of the loan), a solvent guarantor (civil servant, 300 000 FCFA/month salary) and pledged savings of 400 000 FCFA. Combined coverage: 1 200 000 + 400 000 = 1 600 000 > 1 500 000, plus the guarantee.
- Conditions — STRONG. Growing demand for packaged garri, a new market 5 km away, stable input prices, no major competitors. Identified risks: seasonal cassava shortage (Jan–Mar) and poor road access in the rainy season — manageable with stocking and scheduling.

2. Debt Service Coverage Ratio [3 marks].

$$\text{DSCR} = \frac{\text{Net Surplus}}{\text{Monthly Repayment}} = \frac{500\,000}{100\,000} = 5$$

Interpretation: the cooperative generates five times the cash needed to service the debt each month. A DSCR above 1.25–1.5 is generally considered safe; 5 is excellent. Even a severe shock (e.g. surplus halved during the cassava shortage) would leave a DSCR of 2.5, still comfortable. (1 mark for the formula, 1 for the calculation, 1 for the interpretation.)
3. Decision of the Credit Committee [7 marks].
Decision: APPROVE WITH CONDITIONS (the file is fundamentally sound; conditions target the two identified risks: seasonality and collateral formalisation). Three binding covenants:

1. Formalisation of guarantees before disbursement: registration of the chattel mortgage on the grater and press, written and signed guarantee of the Cooperative President, and blocking (pledge) of the 400 000 FCFA savings for the duration of the loan.
2. Controlled disbursement and use of funds: payment made directly to the equipment suppliers against invoices, and construction of the drying shed verified by a Credit Union officer before release of the final tranche — ensuring the 1 500 000 FCFA finances the project and nothing else.
3. Seasonal risk management: obligation to build a cassava stock (or dried chips) before January and to maintain a minimum savings buffer of one monthly instalment (100 000 FCFA) in the pledged account, so that repayments continue during the Jan–Mar shortage and the rainy season.

Examiner expectations: marks are awarded for (i) using the data quantitatively (16.7%, DSCR = 5, coverage 1 600 000), (ii) balanced judgement (recognising the collateral weakness without rejecting a strong file), and (iii) conditions that are specific, binding and linked to the identified risks — vague conditions such as “she must repay well” earn no credit.

11 SUMMARY SHEET

Money and Banking (II)

Bank Balance Sheet
- Assets: uses of funds
- Liabilities: sources

Credit Creation
- Credit multiplier $k = 1/r$
- Leakages limit it

Problems of Banks in Cameroon
- Bad debts, low savings

Central Bank (BEAC)
- Issue, banker to banks
- Monetary policy tools

Microfinance
- MC2, credit unions
- Financial inclusion

Loan Appraisal
- The 5 Cs of credit

Mind map of the chapter: from the commercial bank's balance sheet to loan appraisal by credit unions.

FICHE BILAN **L'essentiel à connaître pour l'examen**

1. Key definitions

- **Balance sheet**: a financial statement showing, at a given date, a bank's assets (uses of funds: cash, loans, investments, premises) and liabilities (sources of funds: deposits, capital, reserves, borrowings). Assets always equal liabilities.
- **Credit creation**: the process by which commercial banks create deposit money by granting loans, since a loan becomes a new deposit elsewhere in the banking system.
- **Central bank**: the apex monetary institution (e.g. **BEAC** for the CEMAC zone) that issues currency, acts as banker to the government and to commercial banks, and conducts monetary policy.
- **Microfinance**: the provision of small-scale financial services (savings, microcredit, micro-insurance) to low-income people excluded from traditional banks; in Cameroon, mainly through **credit unions** and MC^2 institutions.
- **Loan appraisal**: the systematic assessment of a borrower's file before granting credit, based on the 5 Cs: Character, Capacity, Capital, Collateral, Conditions.

2. Formulas and laws to know

- Balance sheet identity: Total Assets = Total Liabilities + Capital.
- Simple credit multiplier: $k = \frac{1}{r}$, where r is the reserve (liquidity) ratio.
- Total deposits created: $\Delta D = \frac{\Delta D_0}{r}$; total credit created: $\Delta C = \Delta D - \Delta D_0$.

- Example: with $r = 20\%$ and an initial deposit of 100 000 FCFA, $k = 5$, total deposits = 500 000 FCFA, credit created = 400 000 FCFA.
- Monetary policy tools of BEAC: **key interest rate**, **reserve requirements**, **open market operations**, refinancing (rediscount) facilities.

3. Methods

- Presenting a balance sheet: classify items into assets (ordered by decreasing liquidity) and liabilities (ordered by increasing exigibility); check the equality of both sides.
- Computing credit creation: identify r, compute $k = 1/r$, multiply the initial deposit by k, then subtract the initial deposit to obtain credit created. Always mention leakages (cash withdrawals, imports, excess reserves) which reduce the multiplier in practice.
- Appraising a loan file (5 Cs): verify the borrower's Character (reputation, repayment history), Capacity (income vs. repayment instalments), Capital (own contribution), Collateral (guarantees), and Conditions (purpose, economic environment).

4. Common mistakes to avoid

- Confusing assets with liabilities: deposits are a liability for the bank (money owed to customers), loans granted are an asset.
- Believing one bank alone creates the full multiplier: credit creation works through the banking system as a whole.
- Forgetting leakages: the formula $k = 1/r$ gives a maximum; real creation is smaller.
- Confusing BEAC (central bank, CEMAC zone) with commercial banks like Afriland, SGBC or BICEC.
- Thinking microfinance charges low interest: rates are often higher because of risk and small loan sizes.

5. GCE tips

- In essay questions on credit creation, always define, give the formula, a numerical illustration, then limits (leakages, demand for loans, BEAC controls).
- For BEAC, structure answers around its functions (issue of currency, banker to banks, banker to the State, guardian of the CFA franc, monetary policy) and mention the COBAC supervisory role.
- Use Cameroonian examples naturally: MC^2 networks, CamCCUL, village savings groups (njangui/tontines) as informal microfinance.
- For problems of banks in Cameroon, cite: low savings and banking rate, bad debts, poor infrastructure, competition from microfinance and mobile money, liquidity concentration in Douala and Yaoundé.

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